From Big Data to "Big Theory"? Evidence from Genome-Wide Association Studies *

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Abstract

Datasets spanning broad portions of a scientific landscape enable a data-driven approach to search, in which empirical patterns rather than prior theory guide which possibilities are investigated. Yet, if theory becomes less necessary for directing search, does this also weaken incentives to develop it? I examine this question in genetics research aimed at identifying new drug targets. I compare candidate gene studies, which select genes based on biological knowledge, with genome-wide association studies (GWAS), which systematically scan the genome for statistical associations with disease. I find that GWAS expand search breadth and increase the variability of scientific outcomes, but with a larger increase in breakthroughs. Importantly, associations uncovered by GWAS attract substantially more follow-on research aimed at explaining their causal basis. Mechanism tests show that these effects arise because GWAS are more likely to uncover empirical anomalies: unexpected associations of varying quality that nonetheless stimulate subsequent theorizing. Together, the findings suggest that reducing theory's role in search can, paradoxically, create new reasons to develop it.

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1 Introduction

Constantin Polychronakos is a leading researcher on the genetic factors behind type 1 diabetes. Pinpointing the genes that carry disease-related mutations is critical because they can reveal targets for therapeutic intervention. Yet, the task is daunting: the human genome is vast, while the time and resources available to investigate it are necessarily limited. Faced with thousands of potential gene-disease combinations, how does Polychronakos decide where to search?

Understanding how Polychronakos searches in the genetic space can offer broader insights into the nature of innovation — not only for individual scientists, but also for organizations and firms. Many technological problems unfold on large combinatorial landscapes defined by prevailing scientific paradigms (Dosi, 1982; Fleming, 2001; Kuhn, 1962). Rather than searching at random, actors concentrate on combinations they expect to be valuable (Felin and Singell, 2025; Kneeland et al., 2020). Actors draw on mental models and cause-and-effect knowledge about how components work together to identify candidates for investigation (Felin et al., 2024; Gavetti and Levinthal, 2000; Sorenson, 2024). Such reasoning makes search efficient by directing effort toward alternatives supported by a plausible theoretical rationale. This theory-driven search is exactly what Polychronakos did in a 2007 study. Building on evidence that the IRF5 gene is implicated in lupus, he hypothesized that IRF5 might also play a role in other autoimmune diseases, leading him to investigate its connection with diabetes (Qu et al., 2007).

The growing availability of big data, however, offers an alternative to theory-driven search. Comprehensive datasets that map large portions of a scientific landscape make it feasible to identify promising combinations to explore without relying on prior theoretical explanations (Agrawal et al., 2024; Ludwig and Mullainathan, 2024; Kim, 2026). I refer to this approach as *data-driven search*, because researchers select candidates for investigation based on patterns in the data, without first having to articulate why they might work. Such methods are increasingly used in settings ranging from the screening of novel pharmaceutical compounds to the development of consumer products (Jayaraj and Gittelman, 2018; Allen and McDonald, 2025). Polychronakos himself adopted data-driven search in a second study in 2007. Rather than selecting genes based on prior biological understanding, he collected genome-wide data from thousands of individuals and looked for genetic variants statistically associated with diabetes. The data identified KIAA0350, a gene of unknown function at the time, as potentially implicated in the disease (Hakonarson et al., 2007).

This change in how research targets are selected creates an epistemic wedge between discovery and understanding (Norvig, 2017). Promising findings can be put to use long before researchers understand their

underlying mechanisms (Evans, 2010). Even in drug development, the U.S. Food and Drug Administration requires that treatments be safe and effective, but not an explanation of how they work. This separation raises concerns that data-driven search may weaken incentives to develop theory (Anderson, 2008; Leonelli, 2014; Mazzocchi, 2015). If empirical patterns can reliably identify new leads, researchers may increasingly devote their efforts to improving data and algorithms rather than theorizing. The success of data-driven search may thus crowd out research aimed at explaining the very relationships it uncovers, even if it ultimately accelerates discovery. The risk is an accumulation of "intellectual debt" as practical advances outpace our understanding of their limits and unintended consequences (Zittrain, 2019).

Empirically examining this concern requires linking how discoveries are made to the research they subsequently stimulate. Yet, in most settings, researchers observe search outcomes but not the process that produced them (Kneeland et al., 2020; Schilling and Green, 2011). In this paper, I address this challenge by studying how scientists investigate the genetic roots of human diseases, a setting in which the same combinatorial landscape can be explored through distinct search strategies. In *candidate gene studies*, researchers draw on prior biological knowledge to target genes expected to matter for a focal disease. In contrast, *genome-wide association studies* (GWAS) screen the entire genome for statistical associations without preselecting genes based on theoretical understanding. Crucially, the design of each study reveals how its gene-disease associations were discovered, allowing me to identify the search process independently of its outcomes. I can then observe how data-driven findings affect subsequent theory development.

I assemble a novel dataset of 369,302 gene-disease associations first reported between 2005, the year of the first GWAS, and 2016. The raw data come from DisGeNET, a comprehensive aggregator of information on genes linked to human diseases. For each gene-disease pair, I identify the first publication reporting the association and all subsequent articles investigating it. I merge these publications with the GWAS Catalog to distinguish associations discovered through genome-wide studies from those identified through candidate gene studies. I measure the scientific value of each association using the DisGeNET Score, a composite index that captures the strength of experimental evidence supporting a gene-disease pair. To characterize follow-on research, I use NIH's PubTator3, which identifies genes and diseases in article abstracts and classifies the relationships reported between them. This allows me to distinguish subsequent studies investigating how a gene is *causally* involved in a disease from those providing additional correlational evidence. The former serves as my proxy for research aimed at developing new theoretical knowledge.

Consider again Polychronakos's two studies on the genetic basis of type 1 diabetes. While the candidate gene study by Qu et al. (2007) explored the association with IRF5 motivated by existing knowledge, the GWAS

by Hakonarson et al. (2007) uncovered an association with KIAA0350 without providing an explanation for why it might be implicated in diabetes. Yet, the latter association proved to be of higher scientific value and attracted 23 follow-on articles, compared with just two for the IRF5-diabetes association. Moreover, most subsequent research focused on elucidating the causal mechanisms linking KIAA0350 to diabetes. That the studies by Qu et al. (2007) and Hakonarson et al. (2007) were conducted by the same principal investigator, in the same year, for the same disease underscores the central point: the search process itself shaped the scope and value of the discovery, as well as the extent of subsequent theorizing generated.

My empirical analyses confirm these patterns. Compared with candidate gene studies, GWAS uncover associations involving historically understudied genes, thus exploring a broader portion of the genetic landscape. GWAS discoveries have higher average DisGeNET Scores; however, they are also more likely to fall in both tails of the quality distribution, indicating greater variance in scientific value. The increase is proportionally larger for the top-tail outcomes, and when restricting to discoveries made by the same researcher, GWAS associations are 36% more likely to fall in the top tail of scientific importance. Importantly, GWAS discoveries stimulate subsequent theorizing rather than crowding it out. They attract substantially more follow-on research, with the increase driven by studies examining the causal mechanisms of the gene-disease association (+141–220% relative to the sample mean). These patterns are robust to stringent fixed effects that compare discoveries made by the same laboratory, for the same disease and year, and even involving the same gene.

A potential concern is that GWAS adoption is a researcher's choice and may be endogenous to the discovery outcomes. Fixed effects mitigate selection on time-invariant disease and researcher characteristics but cannot rule out time-varying shocks that jointly affect method choice and discovery. I therefore introduce an instrumental variable (IV) strategy that exploits the sharp decline in the cost of collecting genetic data. Because genome-wide studies require large cohorts of subjects, this cost decline increased GWAS adoption disproportionately in more prevalent diseases, where large study populations are easier to assemble (Heeney, 2021; Visscher et al., 2017). I capture this differential exposure with a shift-share instrument that interacts annual genotyping costs with disease prevalence measured in 2005. Qualitative evidence and extensive tests indicate that the instrument is both plausible and relevant, with a strong first stage. The resulting 2SLS estimates confirm the robustness of my baseline findings.

Finally, I explore why GWAS elicit more downstream theorizing than candidate gene studies. Drawing on Peirce's notion of *empirical anomalies* (Peirce, 1934; Shi and Evans, 2023), I propose that GWAS are more likely to uncover unexpected findings because they collect data beyond what is needed to test theoretically

motivated hypotheses. Such anomalies can, in turn, raise questions that stimulate theory development (Kuhn, 1962; Locke et al., 2008). I measure anomalousness by assessing how much a new association departs from historical patterns of gene co-occurrence across diseases. The evidence supports this mechanism. First, anomalies are concentrated in both tails of scientific value. Second, GWAS are more likely to introduce such discoveries. Third, the advantage of GWAS in stimulating theorizing increases with the anomalousness of a discovery and is strongest in disease areas with extensive prior genetic knowledge. This runs counter to the intuition that data-driven search matters most where theory provides little guidance. Instead, accumulated knowledge may be what makes empirical surprises more salient as puzzles demanding explanation.

This paper makes three contributions. First, it advances our understanding of how big data changes search (Agrawal et al., 2024; Allen and McDonald, 2025; Hoelzemann et al., 2025). When data cover broad portions of a relevant landscape (Kang, 2025; Kao, 2026; Nagaraj, 2022), actors can identify promising combinations beyond those suggested by existing knowledge, changing both how and where they search. Second, the findings explore the complementarity between data and theory in discovery (Anderson, 2008; Kuhn, 1962; Peirce, 1934; Shi and Evans, 2023). By reducing theory's role in guiding search, data-driven technologies like AI can uncover empirical anomalies that create the need for new explanations (Agrawal et al., 2026). This suggests that the abundance of big data crowds in, rather than crowding out, theorizing. Third, the paper makes a methodological contribution. While modeling innovation over combinatorial landscapes has a long tradition (Levinthal, 1997; March, 1991; Nelson and Winter, 1982), identifying the processes behind observed discoveries remains challenging (Kneeland et al., 2020; Maggitti et al., 2013). I show how deep institutional knowledge allows me to characterize both the strategies actors follow and the landscape of available alternatives, independently of realized outcomes. This makes it possible to examine where different approaches direct search and what the implications are. Broader adoption of this approach could advance the empirical study of search.

2 Conceptual Framework

2.1 The Growing Role of Data in Research

Historians of science have long emphasized advances in scientific instruments as a driver of knowledge advancement (Mokyr, 2002; Rosenberg, 1982; Price, 1983). A key reason is that such tools expand the capacity to generate *data*: they make previously invisible phenomena measurable, rendering those observations usable for analysis (Barbosu and Teodoridis, 2025). For example, model organisms like the OncoMouse enable the collection of data on biological processes that cannot be isolated in humans (Murray et al., 2016). Similarly, motion-sensing sensors collect movement data that was previously difficult to obtain, opening

up new lines of research (Furman and Teodoridis, 2020). By extending the observable portions of specific scientific domains, data-generating tools allow researchers to "tune in and turn on" to new aspects of the natural world (Price, 1983, p. 246).

When data are costly to collect, researchers must carefully choose where to direct empirical investigation (Adner and Levinthal, 2024). In recent decades, however, advances in data-generating tools have made it feasible to map broad portions of scientific and technological landscapes (Barbosu and Teodoridis, 2025). Unlike data collected to investigate specific hypotheses, these mapping efforts aim for systematic coverage of a domain, making observations available across a wide range of potential research targets. In some cases, the mapping of a domain is literal. Satellite programs such as Landsat provide photos of the Earth's surface (Nagaraj, 2022), and the Sloan Digital Sky Survey has mapped large portions of the sky (York et al., 2000). The same logic applies to conceptual spaces. In medicine, the Human Genome Project produced a complete reference genome, while The Cancer Genome Atlas systematically characterized cancer types (Kao, 2026; Kang, 2025). Similarly comprehensive data are now available in fields as diverse as chemistry and economics (Jayaraj and Gittelman, 2018; Nagaraj and Tranchero, 2024).

The ability to map entire knowledge spaces represents more than a quantitative increase in data because it can *qualitatively* change how research unfolds. Rather than collecting data to test pre-specified hypotheses, researchers can examine large datasets for statistical regularities that identify promising candidates for investigation (Ludwig and Mullainathan, 2024; Choudhury et al., 2021; Shrestha et al., 2021). This logic is most applicable in scientific and technological domains where an established paradigm delineates the relevant objects, their feasible recombinations, and the criteria by which progress is evaluated (Dosi, 1982; Kuhn, 1962). In such settings, empirical screening can help address uncertainty about which alternatives to pursue within a well-defined search space (Fleming, 2001).[1] In what follows, I refer to data-driven search as the use of empirical patterns to prioritize leads for investigation.

2.2 Theory and Data as Alternative Ways to Guide Search

Innovation and discovery are often characterized as search over a large combinatorial landscape, where most combinations have low value and a few correspond to high peaks (Langley et al., 1987; Levinthal, 1997). One way to navigate this landscape is through theory-driven search, in which theoretical understanding guides the selection of candidate combinations (Felin and Singell, 2025; Arora and Gambardella, 1994).

[1] This distinguishes the phenomenon studied here from settings of radical uncertainty, including entrepreneurial and market contexts in which the opportunity set is not given and the payoff-relevant landscape is shaped endogenously by actors' choices and strategic interaction (Felin and Zenger, 2017; Felin et al., 2024). Instead, the focus of this paper is on alternative ways to generate salience and prioritize effort allocation within a predefined, albeit vast, opportunity set (Felin and Singell, 2025; Sorenson, 2024).

Cause-and-effect knowledge enables forward-looking inference about which elements may be more likely to solve a focal problem (Felin and Singell, 2025; Sorenson, 2024). Such reasoning can allow researchers to consider possibilities beyond their immediate experience (Gavetti and Levinthal, 2000). It also helps triage ideas by directing data collection toward informative tests (Adner and Levinthal, 2024) or evaluating alternatives through reasoning rather than direct experimentation (Gavetti and Levinthal, 2000). As Popper (1963, p. 36) observed, "theory is a prohibition" and "the more a theory forbids, the better it is": theory makes search more efficient by ruling out outcomes inconsistent with it.

Existing theory acts as a useful filter, providing reasons to prioritize some leads over others. Consistent with this view, prior work documents the advantage of using scientific theory as a "map" for navigating the combinatorial landscape (Fleming and Sorenson, 2004). The same filter, however, carries a cost that is often overlooked: valuable combinations lacking a plausible explanatory rationale can remain unexamined (Chai, 2017). Although theoretical reasoning can reach distant possibilities, researchers can more readily formulate and justify hypotheses where existing knowledge provides a foundation (Knudsen and Levinthal, 2007; Kuhn, 1962). Newton's image of "standing on the shoulders of giants" captures well the inherently cumulative nature of theory-driven search. Search effort may therefore concentrate along established trajectories (Dosi, 1982), making it increasingly path dependent and localized (Levinthal and March, 1993).

Data-driven search can relax this constraint by systematically screening relationships that existing explanations would not have prioritized (Ludwig and Mullainathan, 2024; Shrestha et al., 2021). When data span a broad range of research targets, an empirical pattern can make a hypothesis worth investigating even before researchers understand why it should work (Anderson, 2008). An example is the discovery of halicin, a new antibiotic with an unknown mechanism of action, identified by mining chemical data for statistical associations (Stokes et al., 2020). Signals extracted from the data may thus lower the cost of considering new candidates and increase the likelihood of long jumps in the search landscape (Levinthal, 1997). Data-driven search can also highlight novel opportunities within familiar domains by uncovering new connections among known targets. For instance, associations in genetic data have revealed that cancer types considered anatomically distant share underlying genetic mechanisms (Kang, 2025; Kao, 2026).

Whether data-driven search expands exploration, however, depends on the coverage of the underlying data. Data are often unevenly distributed across a domain, as past research priorities and constraints leave some regions of the landscape poorly observed (Kim, 2026). When observations concentrate on familiar targets or metrics, data-driven search may reinforce existing trajectories, generating a "streetlight effect" (Hoelzemann et al., 2025). In such settings, theory-driven search may be better suited to exploring distant possibilities,

since causal reasoning can generate hypotheses even where empirical evidence is scarce (Camuffo et al., 2024). Conversely, comprehensive mapping efforts can loosen this dependence on past priorities (Nagaraj, 2022), making a wider range of alternatives available for empirical evaluation. Differences in coverage and sampling thus shape which alternatives are evaluated empirically and, in turn, which ones researchers pursue. In March's (1991) terms, data-driven search can foster exploration when data coverage extends outside established research trajectories, but can reinforce exploitation when it largely reflects them.

Note that even when comprehensive data availability leads to broadening exploration, data-driven search need not yield better discoveries (Kaplan and Vakili, 2015). Theory-driven search may identify high-value relationships with fewer attempts, particularly when evaluating candidates requires reasoning beyond the dimensions represented in the data (Felin and Singell, 2025; Allen and McDonald, 2025). Data-driven search may also surface false positives, as apparently promising associations often fail later validation (Tranchero, 2026). At the same time, relaxing theory's upstream filter may uncover exceptional combinations that would otherwise remain unexamined. The net result could ultimately be a more variable distribution of outcomes, with both breakthrough successes and unproductive leads (Girotra et al., 2010). Which search approach yields greater scientific value on average is thus unclear ex ante and likely contingent on the specific context.

2.3 Data-Driven Search and Theory Generation

A discovery can be practically useful even when it lacks a clear explanation of why it works. Accordingly, some have argued that if data-driven search yields valuable outcomes without requiring a prior causal account, incentives to develop new theory weaken (Anderson, 2008; Leonelli, 2014; Mazzocchi, 2015). Consider halicin again. If the goal is to identify a molecule worth pursuing, a data-driven signal of antibacterial activity can guide further investigation even when the mechanism is unclear (Stokes et al., 2020). More generally, causal explanation and predictive performance are distinct objectives, so progress toward one does not require progress toward the other (Norvig, 2017; Shmueli, 2010). This resembles the distinction between studying how x affects y, with $\hat{\beta}$ as the object of interest, and generating predictions $\hat{y}$ to guide a choice (Mullainathan and Spiess, 2017). The concern, then, is one of substitution: when predictive signals can directly identify promising candidates, theory becomes less necessary for deciding where to search. This can reduce the returns to developing causal explanations and shift effort toward improving the data and algorithms that generate those signals.

The "unreasonable effectiveness of data" (Halevy et al., 2009) makes it possible to pursue one promising lead after another without first clarifying why earlier ones worked. Zittrain (2019) describes this accumulation of

poorly understood results as intellectual debt: innovations adopted on the strength of predictive performance create a backlog of unanswered "whys" that must be paid down later through costly validation or failures. This debt is reminiscent of pre-scientific eras of discovery (Arora and Gambardella, 1994). As Mokyr (2002) notes, when theoretical knowledge is limited, research often proceeds through trial and error, with mechanisms clarified only after adoption. Data-driven search reproduces this sequence on a far greater scale, potentially expanding the stock of black-boxed discoveries. The risk of a world rich in discoveries but poor in theory is that practical success outpaces understanding of its limits, leaving boundary conditions and side effects to be discovered later, sometimes only after irreparable harm has occurred (Zittrain, 2019).

While much of the debate emphasizes that data-driven search may crowd out theory, a different possibility follows from the role of surprise in science (Shi and Evans, 2023). Kuhn (1962) argued that scientific paradigms discipline attention by defining what counts as a relevant fact, which can lead researchers to dismiss ideas inconsistent with prevailing assumptions (Chai, 2017). Data-driven methods applied to large datasets can relax this constraint by surfacing unexpected relationships unlikely to appear when focusing solely on pre-specified hypotheses (Agrawal et al., 2026; Choudhury et al., 2021). Some of these findings may constitute *empirical anomalies* in the Peircean sense: observations that sit uneasily with expectations grounded in existing knowledge (Peirce, 1934). As Peirce emphasized, a surprising observation can prompt researchers to formulate an explanatory conjecture that, if true, would make the observation intelligible (Locke et al., 2008). Such anomalies can be theoretically productive precisely because they reveal the limits of prevailing accounts and motivate their extension or revision (Ott and Hannah, 2024).

Notably, it is theory itself that provides the expectations against which empirical anomalies become recognizable (Kuhn, 1962; Peirce, 1934; Kaplan and Vakili, 2015). When prior theory is thin, unexpected associations may not register as puzzles that demand explanation. With stronger priors, departures from expectations are more salient (Camuffo et al., 2024), thereby increasing the impetus to invest in new explanations. This mechanism suggests the possibility that data-driven search stimulates further theorizing especially in domains where prior theory is well developed. Such complementarity, however, is not guaranteed. Even when anomalies are recognized, they compete with other research opportunities for scarce attention (Bikard, 2018; Piezunka and Dahlander, 2015). Data-driven search can intensify this constraint by expanding the supply of empirical leads without improving researchers' capacity to evaluate them (Cornelio et al., 2025; Tranchero, 2026). When attention and experimental resources are limited, anomalies may be pursued superficially or ignored. These considerations make the relationship between data-driven search and theory generation an open question, which I examine empirically in the remainder of the paper.

3 Empirical Setting

3.1 Genetic Research Before Big Data Genomics

Genes are sequences of DNA bases that encode the "instructions" for producing molecules fundamental for an organism's functioning. Mutations in these DNA sequences can alter normal biological processes and thus contribute to disease. Most common conditions, such as diabetes or hypertension, are polygenic: their emergence reflects the combined influence of mutations across multiple genes and environmental factors (Uffelmann et al., 2021). Understanding the genetic roots of such diseases is important because the causative genes can serve as targets for drug intervention. However, identifying such targets is a complex search problem in a search space of over $19,000$ human protein-coding genes.

Scientists have traditionally searched for drug targets using a *candidate gene approach*, which involves three main steps. First, the scientist chooses a disease to study. Second, they hypothesize which genes might contribute to the disease's etiology based on existing knowledge. Third, they test these hypotheses using a case-control design: namely, by examining whether subjects with the condition are more likely to carry mutations in the candidate genes than subjects without it. Importantly, the selection of candidate genes reflects scientists' understanding of which genes are likely to matter for the disease, and why. For example, after the IRF5 gene was associated with lupus, Qu et al. (2007) drew on similarities in the biological pathways underlying autoimmune diseases to hypothesize that the gene might also play a role in diabetes. This led to a candidate gene study that first focused on the IRF5-type 1 diabetes combination (Panel (a) of Figure 1).

Candidate gene studies epitomize a theory-driven model of scientific inquiry. Prior biological knowledge makes genetic search tractable by directing attention toward a limited set of plausible targets. This selectivity economizes on data collection and provides a basis for interpreting the resulting associations. At the same time, requiring a theoretical rationale can leave genes with poorly understood functions outside the set of hypotheses that scientists consider worth testing (Visscher et al., 2017). The focus on known genes has left promising associations unexplored (Stoeger et al., 2018), limiting opportunities to identify therapeutic targets for polygenic diseases (Uffelmann et al., 2021). As such, candidate gene studies illustrate both the guidance provided by existing knowledge and the limits it can impose on search scope.

3.2 The Emergence of GWAS

In the early 2000s, two major developments helped to create an alternative to candidate gene studies. First, the completion of reference genomes such as the Human Genome Project and the HapMap Project provided a comprehensive map of human genes. Second, the cost of collecting large-scale genetic data started to drop

rapidly. Together, these advances paved the way for GWAS as an alternative approach to search for drug targets (Visscher et al., 2017). As the name suggests, GWAS are correlational studies that use genetic data to identify mutations associated with disease across the entire genome.

Similar to candidate gene studies, researchers conducting a GWAS begin by selecting a disease to study and then look for mutations that occur more frequently in individuals with the condition than in those without it. However, the two approaches diverge sharply in how genetic targets are selected (Uffelmann et al., 2021). Rather than focusing on specific gene-disease combinations informed by existing theory or prior evidence, GWAS rely on a fully data-driven approach. Researchers collect DNA samples from large numbers of individuals and systematically test for differences in their genetic makeup across the genome. Genes found to harbor mutations more common in cases than in controls are flagged as potentially implicated in the disease's etiology, regardless of whether their functional role in the disease is already understood. As such, they become candidates for further investigation or pharmaceutical intervention, even in the absence of a clear causal explanation (Visscher et al., 2017).

For example, consider the GWAS conducted by Hakonarson et al. (2007). The researchers analyzed a study population of 563 patients with type 1 diabetes and 1,146 controls without the condition. Using genotyping microarrays, they collected data about the genomes of their study subjects. The comparison between cases and controls highlighted several significant differences: some mutations in genes already known to be involved in diabetes (such as the insulin gene INS), but others in less familiar regions, including the gene KIAA0350 (Panel (b) of Figure 1). At the time, the function of KIAA0350 was unknown, which likely explains why it had been overlooked in candidate gene studies.[2] Since the publication of Hakonarson et al. (2007), the KIAA0350–type 1 diabetes association has proven impactful and has been the focus of multiple follow-up studies. Additional details on this case are provided in Appendix B.

GWAS represent a prime example of data-driven search (Visscher et al., 2017). Unlike candidate gene studies, GWAS generate discoveries purely from patterns that emerge in the data (Panel (c) of Figure 1). By screening genetic variation across the entire genome, they can identify promising associations without requiring a pre-specified functional hypothesis. However, uncovering a new association does not explain the underlying biological mechanism. Even a robust statistical signal can leave open which gene is causally involved, how it affects disease risk, and whether intervening on it would have therapeutic value (Uffelmann et al., 2021). Moreover, achieving sufficient statistical power across the genome requires large sample sizes,

[2]Even the gene's name showed its limited characterization at the time, because KIAA0350 was a temporary placeholder assigned upon its discovery. The gene was renamed CLEC16A ("C-type lectin domain containing 16A") when its function became clearer.

leading some skeptics to dismiss GWAS as a costly detour in the quest for genetic insight[3]. These limitations leave an important role for theory-driven research, explaining why candidate gene approaches remain in use and questions persist about the value of GWAS.

4 Measurement Strategy and Data

4.1 Defining the Search Landscape: Gene-Disease Combinations

While search is often conceptualized as occurring within a combinatorial landscape (Fleming, 2001; Levinthal, 1997; Schilling and Green, 2011), such landscapes are typically difficult to study empirically. This paper leverages the setting of genetic research to address that challenge. A central goal in genetics is to identify which genes could serve as drug targets for a given disease. In this context, genes and diseases constitute the relevant components of the search problem. Any gene can, in principle, be tied to any disease, creating a space of millions of potential gene-disease combinations (see Appendix C.1 for an illustration). Importantly, starting in the early 2000s, reference genome maps have provided a common set of genes that can be examined across disease areas (Jayaraj and Gittelman, 2018). This allows the landscape of potential gene-disease combinations to be characterized separately from how search unfolds over it.

I obtain data on links between human diseases and genetic mutations from DisGeNET (Piñero et al., 2020), an aggregator of scientific sources considered one of the most comprehensive repositories of genetic knowledge. The database compiles gene-disease associations (GDAs) from specialized sources, including curated datasets and PubMed-indexed publications. The units of observation are gene-disease pairs introduced in scientific articles published between 2005 and 2016. For each pair, I assemble the first publication reporting it and all subsequent articles investigating it. I focus on associations that link a protein-coding gene to a disease, syndrome, or abnormality with health implications. The final dataset includes 369,302 newly reported gene-disease associations, encompassing a total of 14,136 genes and 9,863 diseases.[4] Bibliometric information for papers introducing novel gene-disease associations is taken from NIH's iCite data. To identify the principal investigator (PI), I use data on the last author of each publication from the Author-ity database (Torvik and Smalheiser, 2021).

[3]Over time, the decrease in genotyping costs led to a greater diffusion of GWAS. In Section 5.3, I exploit the cost reduction as an instrument for the adoption of a GWAS approach. For a critical discussion of early GWAS costs, see: *www.blog.goldenhelix.com/have-we-wasted-7-years-and-100-million-dollars-on-gwas*.

[4]Since the analysis focuses on comparing GWAS and candidate gene studies, the sample is restricted to new gene-disease combinations introduced on or after 2005, the year of the first GWAS. As a result, the number of genes in the dataset is lower than the total number of human genes, as some were not implicated in any disease during the sample period.

4.2 Data-Driven and Theory-Driven Search in Genetics

Empirically investigating how alternative search strategies shape outcomes depends on linking the search process to what it produces. This is challenging because researchers typically observe outcomes but not the search process that generated them (Kneeland et al., 2020; Maggitti et al., 2013). A common workaround is to infer search strategies from observable features of outputs. For example, the breadth of a paper's references or the diversity of a patent's USPC classes have been used as indicators of how inventors searched (Fleming, 2001; Schilling and Green, 2011). The problem is that these proxies are inherently ex post and can conflate the search strategy with its realized outcome. Empirical analysis, therefore, requires a setting in which the search strategy can be observed independently of the outcomes it produces.

Genetic research makes this possible because the study design itself reveals whether search was theory-driven or data-driven. Candidate gene studies vary in the extent of genetic targeting, ranging from testing the association between a single gene and a disease (as in the paper by Qu et al., 2007) to examining broader regions that may contain several genes. GWAS, by contrast, involve no genetic targeting at all. I identify these studies using the GWAS Catalog, a manually curated database of studies adopting a genome-wide approach (MacArthur et al., 2017). Crucially, an explicit criterion for inclusion in the Catalog is the absence of ex ante gene selection. I can therefore identify the search method behind each gene-disease association without having to infer it from the characteristics of the discovery. In total, the DisGeNET data include 8,655 gene-disease associations first introduced by 1,250 distinct GWAS, while the remainder come from studies involving genetic targeting.[5]

4.3 Characteristics of Gene-Disease Combinations

The empirical section examines how gene-disease associations introduced by GWAS differ from those introduced by candidate gene studies. Specifically, I compare their novelty, scientific quality, follow-on research, and anomalousness relative to prior genetic knowledge.

A. Novelty of combination: I use two alternative dependent variables to capture discoveries involving genes that had received scant attention before the emergence of GWAS. The first is a dummy variable that equals one if the gene in the newly introduced gene-disease association had never been linked to any disease.

[5] Studies not recorded in the Catalog include a range of targeted and hybrid designs, which I group under the candidate gene category. Any genome-wide study not included in the Catalog would be coded in this comparison group, which would attenuate my estimates. Also, note that the GWAS Catalog records all statistically reported gene-disease associations in each GWAS paper, even when those associations are not novel. By construction, GWAS studies often re-identify previously documented associations, and about 67% of the associations reported in the Catalog are rediscoveries (Appendix Figure A.2). As a result, my data are unlikely to be biased by strategic reporting or file drawer issues (Appendix A.3). The GWAS Catalog applies a Bonferroni correction and reports as significant only associations that meet a stringent threshold (p-value $< 1.0 \times 10^{-5}$).

Returning to the example from the introduction, the KIAA0350 gene is coded as underexplored because it had not been associated with any disease before the study by Hakonarson et al. (2007). The second proxy is the gene's discovery date, since genes mapped more recently tend to be less understood (Stoeger et al., 2018). Accordingly, I test whether GWAS are more likely to introduce combinations involving genes discovered after the first draft of the human genome was released in 2000. For example, the KIAA0350 gene was discovered only in 2002. Both dependent variables are dichotomous, so the OLS specifications can be interpreted as linear probability models. I replicate the analyses using continuous versions of these variables in the Appendix.

B. Quality of combination: Researchers typically rely on paper-to-paper citations to proxy for scientific quality, but that approach is difficult to apply in my setting because publications in genetics often report multiple discoveries. For example, the GWAS by Hakonarson et al. (2007) reported associations between type 1 diabetes and five genes (KIAA0350, INS, COL1A2, LPHN2, and PTPN22), making it difficult to assign a clear share of the article's citations to each individual discovery.[6] To address this issue, I use the DisGeNET Score, a unique metric available at the association level (Piñero et al., 2020). The Score is a composite index that reflects the underlying scientific value of each gene-disease pair, constructed from independent experimental evidence.[7] For example, the KIAA0350–type 1 diabetes association has a Score of 0.46, placing it in the 95th percentile for scientific quality, compared with 0.03 for IRF5–type 1 diabetes. Appendix C.2 further details the construction of the DisGeNET Score. Supporting its validity, articles introducing combinations with higher Scores receive more citations (Appendix Table C.1), and combinations with higher Scores are associated with greater impact on drug discovery (Appendix Table C.2). This metric allows me to assess how search methods shape the quality of discoveries, including their likelihood of falling in the top and bottom tails of scientific quality.

C. Follow-on research on the combination: To examine whether data-driven discoveries attract research aimed at theory-building, I need a measure of downstream work that directly investigates a specific finding. I construct this measure by identifying subsequent papers that mention the same gene and disease in their abstracts, regardless of whether they cite the original study. For example, following their discovery in 2007, the KIAA0350–type 1 diabetes association attracted 23 follow-on papers, compared with just 2 for IRF5–type

[6]Paper-to-paper citations also capture attention allocation (Bikard, 2018). GWAS papers are, on average, much more cited than candidate gene studies ($Cites_{GWAS}$ = 170 vs. $Cites_{Candidate\ Gene\ Study}$ = 41). These citations can reflect the novelty of the genome-wide approach itself, including reviews and methodological debates, without indicating higher-quality discoveries.

[7]Researchers and pharmaceutical firms use the DisGeNET Score to prioritize targets for drug discovery. One component of the DisGeNET Score increases with the number of supporting publications and may partly reflect the allocation of follow-on research effort. In the appendix, I assess this concern by recomputing the Score and, separately, controlling for the number of supporting publications and non-literature sources. The results are unaffected by these adjustments (Appendix Table C.3).

1 diabetes. Several of these studies do not cite the originating paper and would be missed by citation-based measures (Arts et al., 2025). To characterize the nature of this follow-on work, I use NIH's PubTator3 (Wei et al., 2024), which uses AI to classify the type of relationships described in article abstracts, such as a gene inhibiting a disease or being associated with it. I use these relationship labels to separately count follow-on articles that report causal mechanisms and those that add correlational evidence for the focal gene-disease pair. The former serves as a proxy for research aimed at developing a theoretical understanding. Supporting the face validity of the measure, Appendix Table D.1 shows that clinical trial papers are more likely to be classified as reporting causal relationships. Returning to the example, 48% of the papers that build on the KIAA0350–type 1 diabetes association focus on its causal drivers.

D. Anomalousness of the combination: Adopting the notion of Peirce (1934), I define empirical anomalies as new gene-disease associations that are surprising in light of existing knowledge (Shi and Evans, 2023). To measure this surprise, I exploit the fact that different genes contribute to the same disease by disrupting a shared biological process (Visscher et al., 2017). This means that prior gene-disease associations can provide clues about which additional genes might be implicated in a disease. Specifically, a new association is less surprising if the newly implicated gene has previously been linked to other diseases alongside genes already associated with the focal disease. Let $G_{d,t-1}$ denote the set of genes associated with disease d before year t, and let $f_{gh,t-1}$ denote the number of diseases with which genes g and h had both been associated before that year. I define the Anomaly Index for a newly reported association as:

$$\text{Anomaly Index}_{gd,t} = \frac{1}{1 + \sum_{h \in G_{d,t-1}} f_{gh,t-1}}. \tag{1}$$

The Index is higher when the focal gene has rarely co-occurred with genes already associated with the disease, thereby capturing discoveries that depart from known patterns (Appendix E). Two validation exercises support the measure's face validity. Using lexical cues from paper abstracts, Appendix Table E.1 shows that discoveries with higher Anomaly Index values are more likely to be described as "novel," but not with other generic hype terms. Appendix Table E.2 shows that these discoveries are more likely to appear in papers tagged as "new findings" by expert reviewers on the Faculty Opinions platform.

4.4 Summary Statistics

Table 1 lists the key variables along with descriptive statistics for the sample used in the analysis. Panel A presents statistics at the publication level, focusing on papers that introduce new gene-disease combinations

during the study period. Compared with candidate gene papers, GWAS tend to report more associations and span more genes, consistent with their untargeted nature. Panel B provides summary statistics on new gene-disease combinations. GWAS and candidate gene studies introduce new discoveries with similar average DisGeNET Scores. However, GWAS associations are more likely to appear in both the bottom and top of the Score distribution, indicating greater variability in their scientific value. Combinations introduced by GWAS receive more follow-on research, much of which focuses on clarifying the causal relationship between the gene and the disease, and have higher Anomaly Index values.

5 Results

5.1 How Big Data Changes Genetic Search

In this section, I begin by examining how gene-disease associations discovered through a data-driven search process differ from those introduced via a theory-driven one. Figure 2 provides an intuitive visualization of the combinatorial space of potential gene-disease pairs, showing where new discoveries in my data occur. Panel (a) shows that candidate gene studies largely build on existing knowledge, thus concentrating discoveries on well-studied genes. Panel (b) reveals a sharp contrast: GWAS associations span a much broader portion of the genetic landscape, including genes overlooked by earlier research. Importantly, GWAS continue to focus on diseases that have historically attracted substantial scientific attention, since disease selection remains with the PI under both approaches. This pattern indicates that the increased breadth of discovery reflects the removal of discretion over gene targeting, rather than a shift in researchers' disease focus.

To quantify these differences, I estimate the following cross-sectional OLS model at the gene-disease association level:

$$(Features\ of\ gene\text{-}disease\ combination)_{i,j} = \alpha + \beta GWAS_{i,j} + \mathbf{Z}'_{i,j}\gamma + \epsilon_{i,j} \tag{2}$$

where the characteristics of the gene-disease pair (i, j) are regressed on the indicator variable $GWAS_{i,j}$ for whether the association was discovered by GWAS rather than a candidate gene study. The vector $\mathbf{Z}'_{i,j}$ contains fixed effects for targeted disease, PI, and year of first publication of the combination. With PI fixed effects, the coefficient of interest is identified by comparing discoveries introduced by the same scientist using the two search approaches, holding constant time-invariant researcher characteristics. If data-driven search systematically leads to different outcomes, I should observe a statistically significant estimate of β. All specifications cluster standard errors two ways at the gene and the disease level.[8]

[8]These estimates are descriptive and may still be affected by unobserved, time-varying confounders. Section 5.3 discusses this concern and addresses it using an instrumental variable strategy to mitigate endogeneity in the choice of research method.

Table 2 reports the results. Column 1 shows that GWAS increase by 26 percentage points the likelihood that a newly discovered association involves a gene never linked to any disease before 2005, relative to a sample mean of 13 percentage points. Column 3 corroborates this finding using the gene's discovery date as an alternative proxy for prior scientific attention. Even when comparing discoveries made by the same laboratory, GWAS are 106–148% more likely to introduce associations involving historically understudied genes, relative to the sample means in Table 1. Appendix Table G.1 confirms these patterns using continuous versions of the dependent variables. On average, GWAS discoveries involve genes that received 29–51% fewer publications before 2005 and were discovered 2.1–3 years later.

These findings speak to a long-standing puzzle in genetics: why research remains concentrated on a small subset of genes despite potentially large gains from broader exploration (Stoeger et al., 2018). My results point to a mechanism inherent to candidate gene studies. When prior knowledge is required to formulate hypotheses about which genes to investigate, attention naturally concentrates on familiar targets whose biological functions are better understood. GWAS, by contrast, relax this filter, allowing less familiar genes to emerge as candidates for investigation (Visscher et al., 2017). The results suggest that data-driven search can loosen the path dependence of theory-driven search.

5.2 The Impact of GWAS on Scientific Discovery

While data-driven search appears to broaden the scope of search, there is no guarantee that the resulting discoveries are more valuable or generative of new understanding. If anything, one might reasonably expect the opposite. Scientists should already be focusing on what they consider the most promising areas of the scientific landscape, making it less likely that a data mining exercise will yield better discoveries. I examine this question in Table 3. Panel A shows that gene-disease combinations introduced by GWAS have, on average, a DisGeNET Score that is 20–21% higher. Interestingly, this effect is driven by a thickening of both tails: GWAS combinations are more likely to be either below-median discoveries or high-quality ones. This pattern is consistent with the idea that data-driven search increases outcome variability because it lacks theoretical filtering. Notably, the increase in top-tail discoveries is proportionally larger, which explains the overall rise in average DisGeNET Score for GWAS-introduced combinations.[9]

Panel B examines the downstream scientific consequences. A natural concern is that a data-driven approach could crowd out explanation, leading to findings that are applied without stimulating work on their underlying

[9]These results hold under alternative thresholds for defining the tails of the DisGeNET Score distribution (Appendix Tables G.2 and G.3). Further supporting the results, the coefficients increase proportionally with more stringent definitions of failure or breakthrough, confirming that GWAS are especially powerful at uncovering outlier pairs in both quality tails.

mechanisms. Instead, my estimates point to the opposite. Gene–disease associations introduced by GWAS attract substantially more follow-on work. A GWAS association is followed on average by 0.82 additional papers in specifications without PI fixed effects (+140% over the sample mean) and by 0.54 additional papers when comparing discoveries made by the same investigator (+93% over the sample mean). Importantly, this increase is driven by research aimed at understanding mechanisms. Follow-on articles probing the causal determinants of the gene–disease link rise by 0.29–0.45, a 141–220% increase over the sample mean, whereas the estimates for correlational follow-on work are close to zero and, depending on the specification, not statistically significant.[10] In total, 95% of causal follow-on papers on GWAS discoveries originate outside the discovering lab, indicating broader uptake across the field. Together, these findings suggest that GWAS discoveries crowd in subsequent theorizing.

Two additional analyses help rule out potential confounders. First, the results are not driven by lab-specific changes in disease focus. Appendix Table G.4 shows that the estimates remain similar with more stringent fixed effects, comparing discoveries for the same disease in the same year, and even within the same laboratory for the same disease and year. Second, the results are not simply a consequence of GWAS drawing from a different pool of genes. When I include gene fixed effects, which compare GWAS and candidate gene discoveries involving the same gene, GWAS associations still have higher DisGeNET Scores and attract more follow-on work (Appendix Table G.5). However, the increased likelihood of bottom-tail discoveries disappears, suggesting that this result partly reflects GWAS uncovering lower-potential genes that theory-driven search tends to (correctly) screen out. Overall, the higher scientific value of GWAS discoveries is not simply driven by differences in the diseases or genes studied.

5.3 Instrumental Variable Estimates

The adoption of GWAS is a researcher's choice and may therefore be endogenous to the outcomes documented in the previous section. A first concern is selection. Researchers who are systematically better or worse at identifying breakthroughs may differentially adopt GWAS, thereby biasing the estimates.[11] Similarly, GWAS could be deployed in disease areas with more "low-hanging fruits," which would bias estimates upward, but it could also be adopted precisely where theory-driven search is running into diminishing returns, which would bias estimates downward. To mitigate these concerns, each table includes specifications with disease and

[10] One concern is that GWAS discoveries may attract more follow-on research because they appear in more prestigious journals. Appendix Table G.6 shows that the results are robust to controlling for the prestige of the journal in which each association was first reported, using alternative measures based on SCImago journal rankings.

[11] The direction of this bias is ambiguous. If scientists adopt GWAS after struggling to identify valuable combinations through theory-driven search, the bias would be downward. If, instead, the most capable scientists are early adopters, the bias would be upward. Appendix Figure G.1 allays these concerns by showing that, prior to their first GWAS paper, eventual adopters do not introduce gene-disease associations of systematically different scientific quality.

PI fixed effects, where coefficients are identified from within-disease and within-laboratory comparisons. However, a more serious issue would be time-varying drivers of GWAS adoption. Unobserved shocks, such as disease-specific increases in funding, could both increase the likelihood of switching to GWAS and affect the subsequent outcomes. Fixed effect estimation does not eliminate this type of confounding.

To address these concerns, I adopt an instrumental variable (IV) strategy that isolates plausibly exogenous variation in GWAS adoption. The IV exploits a technology-driven shock to the feasibility of genome-wide search. Panel (a) of Figure 3 shows the sharp decline in the cost of collecting genetic data beginning in the mid-2000s, following the diffusion of low-cost microarray technologies (Visscher et al., 2017). This drop in costs represents an exogenous supply-side shift that increases the attractiveness of a genome-wide approach relative to targeted candidate-gene studies.[12] Importantly, the same cost decline did not translate into uniform adoption across disease areas. GWAS require large cohorts that are easier to recruit for more prevalent diseases, and funding agencies prioritized those same diseases given their public-health salience. As a result, the cost decline generated larger increases in GWAS use precisely in disease areas with higher baseline prevalence in the population (Panel (b) of Figure 3).

To operationalize this intuition, I construct a shift-share instrument that interacts annual sequencing costs with the number of individuals living with a given disease before GWAS emerged. I obtain yearly cost data from the NIH and measure disease prevalence in 2005 using public data from the Global Burden of Disease study and Orphanet (details in Appendix F). Prevalence information is available for 1,724 diseases, which defines my IV sample.[13] The instrument varies at the disease-year level and predicts GWAS adoption through differential exposure to the common decline in genotyping costs:

$$\text{Genotyping cost shift-share}_{j,t} \;=\; \text{Genotyping Cost}_{t} \times \text{Disease Prevalence}_{j,2005}. \tag{3}$$

I then estimate the first stage as the likelihood that a combination (i, j) is discovered in year t using a GWAS rather than a candidate gene study:

$$\text{GWAS}_{i,j} \;=\; \pi_0 + \pi_1 \,\text{Genotyping cost shift-share}_{j,t} + \mathbf{Z}'_{i,j}\pi_3 + u_{i,j}, \tag{4}$$

where i indexes genes, j indexes diseases, and $\mathbf{Z}_{i,j}$ includes the baseline controls and fixed effects from the main specification. The second stage replaces the endogenous GWAS indicator with its fitted value from the

[12] I thank an anonymous referee for this helpful suggestion.

[13] Appendix Table G.7 shows that the baseline OLS findings on scientific value and follow-on research are replicated when estimated on the smaller IV sample.

first stage:

$$(Features\ of\ gene\text{-}disease\ combination)_{i,j} \;=\; \alpha + \beta_{2SLS}\,\widehat{\text{GWAS}}_{i,j} + \mathbf{Z}'_{i,j}\gamma + \varepsilon_{i,j}, \tag{5}$$

where β_{2SLS} is the IV estimate identified from differential changes in GWAS adoption across diseases as genotyping costs fall. Because the instrument has a shift-share structure, Appendix F.4 reports the diagnostics recommended by Borusyak et al. (2025). In particular, I show that diseases with higher versus lower baseline prevalence exhibit similar pre-GWAS trends in the outcomes and that my estimates are stable in "leave-one-disease-out" specifications.

Table 4 reports 2SLS estimates for the diseases with available prevalence data. The first column of each panel presents the first stage. As expected, higher instrument values, which correspond to years with higher genotyping costs, predict a lower probability that a newly reported association is discovered via GWAS ($\hat{\pi}_1 = -0.0104$, *p-value* $< .001$). The first stage is strong despite the smaller sample, with Cragg-Donald F statistics $= 53.6$ and Kleibergen-Paap F statistics $= 19.5$. The remaining columns replicate the outcome specifications from Table 3. Across panels, the 2SLS estimates match the OLS results in sign and are substantially larger.[14] The larger magnitudes are consistent with the IV estimates capturing a local average treatment effect (LATE) for discoveries in disease areas where declining genotyping costs induced a shift toward GWAS. Overall, the IV results strengthen a causal interpretation of the baseline findings: GWAS introduce gene-disease associations of higher scientific value and stimulate more follow-on research.

5.4 Mechanisms

The findings from the previous sections show that the way scientists search shapes the features of their discoveries. What remains to be explained is why a data-driven approach like GWAS appears particularly generative of follow-on theorizing. A potential mechanism, suggested by the philosophy of science, is the uncovering of empirical anomalies (Chai, 2017; Kuhn, 1962; Peirce, 1934). Unlike candidate gene studies, GWAS perform systematic genome-wide screening, increasing the likelihood of uncovering surprising relationships that cannot be deduced from existing knowledge. This could help explain both of my earlier findings. On the

[14] The IV estimates also confirm that GWAS are more likely to uncover associations involving understudied genes (Appendix Table G.8). Across outcomes, the only pattern that does not survive is the increase in low-quality combinations. One speculative explanation is differential selective reporting. Fanelli (2010) shows that studies testing multiple hypotheses are more likely to report null results, since unfavorable findings are easier to publish when they can be bundled alongside significant results. Indeed, bottom-tail associations are uncommon in candidate gene papers reporting a single association (9.6%), but less so in those reporting multiple associations (25.2%). In contrast, GWAS are less conducive to suppressing low-value findings because researchers typically report the full set of associations, resulting in a higher rate of low-quality findings (35.1%). If low-value single-hypothesis results are disproportionately hidden in the file drawer, OLS will exaggerate the bottom-tail gap between GWAS and candidate gene studies, which could possibly explain why this effect attenuates in the IV specifications.

one hand, anomalies may be expected to encompass both valuable discoveries and spurious associations; on the other hand, unexpected findings can expose gaps in existing knowledge and motivate research to develop new causal explanations (Mullainathan and Rambachan, 2024). I examine these possibilities below.

First, I find that anomalous gene-disease associations are concentrated in both tails of the DisGeNET Score distribution (Appendix Table G.9, Panel A). GWAS are, in turn, substantially more likely to uncover anomalous associations, even after accounting for their tendency to involve less-studied genes (Appendix Table G.10). This helps explain the greater variability in the scientific value of GWAS discoveries. An increase in variability may be particularly valuable in disease areas with extensive prior genetic knowledge. Prior work shows that more accurate evaluation of alternatives can paradoxically constrain search, trapping actors at local optima (Knudsen and Levinthal, 2007). Similarly, extensive genetic knowledge may make theory-driven search too selective, excessively concentrating attention on known genetic mechanisms. Consistent with this possibility, Appendix Figure G.2 shows that GWAS yield proportionally more high-quality discoveries in disease areas with extensive prior genetic knowledge.

Second, I examine whether anomalies help explain the greater theorizing stimulated by GWAS. Anomalous discoveries do not, on average, attract more follow-on research (Appendix Table G.9, Panel B). This aligns with Kuhn's view that scientists' first instinct is to ignore anomalies rather than treat them as puzzles demanding their attention (Chai, 2017; Kuhn, 1962). Yet, Appendix Figure G.3 reveals an important difference between search approaches. GWAS associations receive more follow-on research on causal mechanisms, an advantage that grows for more anomalous discoveries. One possibility is that candidate gene discoveries, even when anomalous, may have a plausible biological rationale, whereas GWAS leave more for subsequent researchers to understand. Moreover, systematic genome-wide screening and stringent statistical thresholds may lend credibility to these findings, encouraging scientists to investigate rather than dismiss them.

Finally, I explore when anomalous GWAS discoveries attract the strongest response. Prior knowledge could matter in opposing ways. In well-studied disease areas, scientists may be more inclined to dismiss unexpected associations that do not fit existing explanations. Yet accumulated knowledge may also make such findings more meaningful as puzzles demanding explanation (Ott and Hannah, 2024), while providing a foundation for investigating their causal mechanisms. These areas may also offer greater expected returns to further research, strengthening incentives to pursue unexpected findings. Figure 4 is consistent with this second possibility. The GWAS advantage in attracting causal follow-on research is largest for anomalous discoveries in disease areas with extensive prior genetic knowledge, and smaller when discoveries are less surprising or

prior knowledge is limited. Although these results are descriptive, they suggest that data-driven search may be particularly generative of theorizing in areas where substantial knowledge has already accumulated.

6 The Dynamic Effects of Data-Driven Discoveries

The preceding results show that GWAS discoveries attract more follow-on research, especially work aimed at clarifying causal mechanisms. This section examines whether this response reflects durable theory building or reflects a short-lived burst of attention, and whether the additional research is scientifically valuable. To study these dynamics in a difference-in-differences setup, I construct a gene-disease-year panel using DisGeNET. Each observation records the number of articles examining a given gene-disease pair in year t, excluding the original discovery paper. Summary statistics are reported in Appendix Table G.11.

Figure 5 plots the average number of publications following the introduction of a new gene-disease association. GWAS-introduced associations attract substantially more follow-on research than candidate gene discoveries, and this difference persists over time. To examine whether this pattern holds after accounting for differences across associations and changes in research attention over time, I estimate the following panel specification:

$$(Papers\ on\ gene\text{-}disease\ combination)_{i,j,t} = \alpha + \beta_1 Post_{i,j,t} + \beta_2 (Post_{i,j,t} \times GWAS_{i,j}) + \lambda_{i,j} + \delta_{j,t} + \omega_{i,t} + \epsilon_{i,j,t} \quad (6)$$

where $Papers_{i,j,t}$ is the number of articles examining gene i and disease j in year t, excluding the original discovery paper. $Post_{i,j,t}$ equals one in the years following the initial publication of the gene-disease association, and $GWAS_{i,j}$ equals one if the association was first reported in a GWAS. Gene-disease pair fixed effects ($\lambda_{i,j}$) account for time-invariant differences in the research potential of specific associations. Disease-year fixed effects ($\delta_{j,t}$) and gene-year fixed effects ($\omega_{i,t}$) account for changes in research attention to individual diseases and genes over time. The coefficient β_2 captures the differential change in follow-on research associated with GWAS discoveries relative to candidate gene discoveries. Standard errors are clustered at the gene and disease levels.

Appendix Table G.12 reports the main results. Consistent with the cross-sectional evidence, GWAS-introduced associations attract more follow-on papers per year than candidate gene discoveries, with the difference concentrated in research examining causal mechanisms. The event study estimates in Figure 6 show that this increase emerges after discovery and remains sizable even eight years later (Panel a), with no comparable increase in correlational work (Panel b). Table 5 assesses the impact of this follow-on work by re-estimating the panel specification with annual article counts weighted by scientific citations, citations

from clinical trials, and the SCImago Journal Rank (SJR) of the publication outlet. The estimates are positive across all three measures, indicating that the additional research has a larger scientific impact. Together, these results suggest that the effects of GWAS discoveries extend beyond an initial burst of attention, generating a sustained and valuable stream of subsequent theorizing.

7 Discussion

This paper examines how data-driven search shapes scientific discovery and theory generation by comparing GWAS with candidate gene studies. GWAS broaden the genetic landscape that researchers empirically examine, uncovering new gene–disease links that are more likely to implicate previously underexplored genes. The resulting discoveries exhibit greater variability in quality, with a pronounced right-tail increase: breakthroughs outpace low-value findings, raising the average scientific yield. Such discoveries also attract substantially more follow-on research aimed at clarifying causal mechanisms. An IV strategy exploiting the decline in genotyping costs confirms that the changes in discovery outcomes are due to the different search approach adopted. Mechanism evidence suggests that GWAS stimulate theorizing by uncovering empirical anomalies that challenge theoretical expectations. Consistent with this interpretation, anomalous GWAS findings elicit the strongest follow-on response in areas with more accumulated genetic knowledge.

My results underscore the nuanced ways in which alternative search strategies shape theory development. Theory-driven search makes exploration efficient by prioritizing combinations supported by existing knowledge (Fleming and Sorenson, 2004; Felin and Singell, 2025), but this reliance on prior work can also leave promising paths unexplored (Levinthal and March, 1993; Knudsen and Levinthal, 2007). By contrast, data-driven search can uncover relationships that researchers would not have prioritized on theoretical grounds, providing an empirical basis for long jumps into less explored parts of the landscape (Levinthal, 1997). Empirical surprises can, in turn, raise questions that motivate new explanations (Kuhn, 1962; Locke et al., 2008; Peirce, 1934). Thus, reducing reliance on theory to guide discovery can ultimately create new opportunities for its development. To paraphrase Louis Pasteur, "chance favors the curious mind," and data-driven patterns can be exactly what sparks that curiosity.

Nevertheless, greater data availability need not diminish the value of theory-driven search. Theoretical reasoning can open new directions beyond those represented in the data (Camuffo et al., 2024; Felin et al., 2024). This makes targeted investigation especially valuable when observations are costly to obtain (Adner and Levinthal, 2024) or the available data are biased (Hoelzemann et al., 2025). As the opening example illustrates, the same scientist can fruitfully draw on both approaches; however, in my data, most research

on the causal mechanisms underlying GWAS associations does not come from the researcher making the discovery. This suggests a division of labor in which one researcher's data-driven discovery becomes another's theoretical puzzle. For research organizations, investments in data capabilities may therefore be more productive when accompanied by the expertise needed to evaluate and investigate the resulting leads (Tranchero, 2026). Such complementarity also helps explain why access to more data does not diminish the value of firms' scientific capabilities and their longstanding links with academia (Arora et al., 2025; Rosenberg and Nelson, 1994).

While human genetics provides a clean setting for this study, my findings have boundary conditions. The results speak most directly to scientific and technological search in combinatorial problem spaces, where established frameworks define the objects of search and their feasible recombinations, making systematic data collection possible (Dosi, 1982; Kuhn, 1962; Felin and Singell, 2025). Under these conditions, data-driven screening can help actors address combinatorial uncertainty by expanding "offline" investigation of alternatives (Gavetti and Levinthal, 2000). By contrast, many entrepreneurial and market settings involve opportunity sets that actors help construct through their beliefs and choices (Camuffo et al., 2024; Felin and Zenger, 2017; Ott and Hannah, 2024). In such cases, the landscape is endogenous to firms' actions and strategic interaction, limiting any notion of a "ground truth" that data can simply reveal. Future research should examine how these differences shape the relationship between data-driven search and theory generation.

In conclusion, the history of science and technology offers many examples of anomalous empirical observations that led to theoretical advances (Kuhn, 1962; Price, 1983). For instance, Galileo's telescope enabled observations that fueled the Copernican Revolution, and Rosalind Franklin's X-ray diffraction images helped establish the structure of DNA. What makes the findings of the present paper especially timely is that AI and related data technologies are expanding the scale and speed at which empirical anomalies can be surfaced (Agrawal et al., 2026; Conti and Messinese, 2024; Mullainathan and Rambachan, 2024). By lowering the cost of screening large quantities of data, AI may loosen the path dependence that accompanies cumulative learning and direct attention toward relationships that prove generative of new theorizing. In this light, a more data-driven search process may not mark the "end of theory," as Anderson (2008) once proclaimed, but rather the beginning of an era in which big data leads to "big theory."

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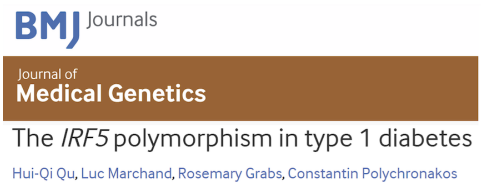
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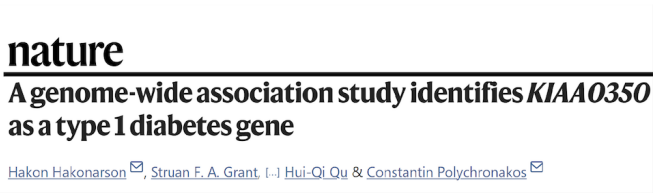
8 Figures and Tables

Figure 1: Scientists search for new gene-disease associations with candidate gene studies or with GWAS.

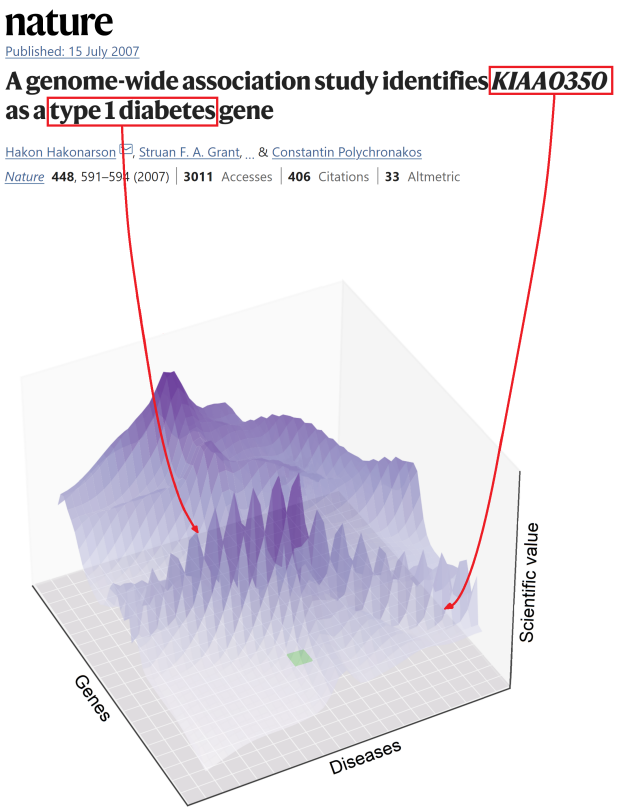
(a) *Example of candidate gene study*



(b) *Example of GWAS*



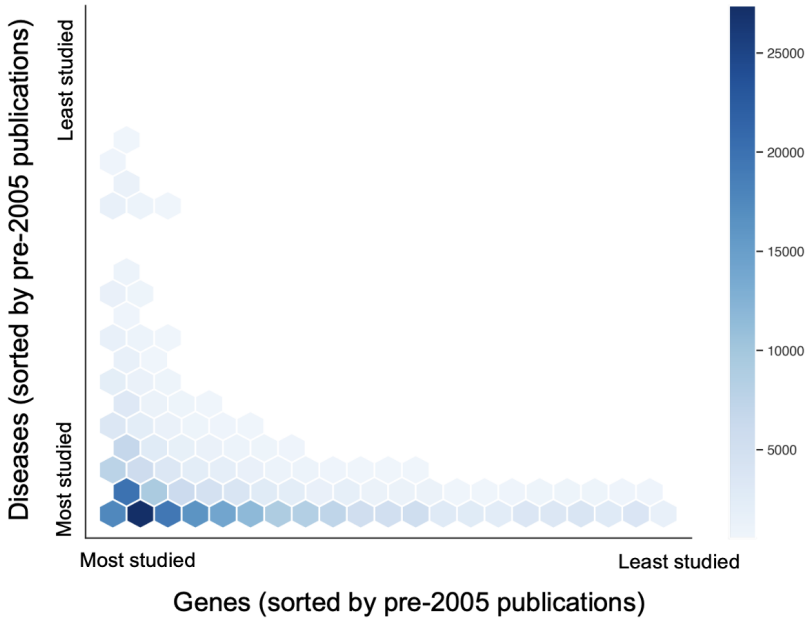
(c) *GWAS as data-driven search on a landscape of gene-disease combinations*



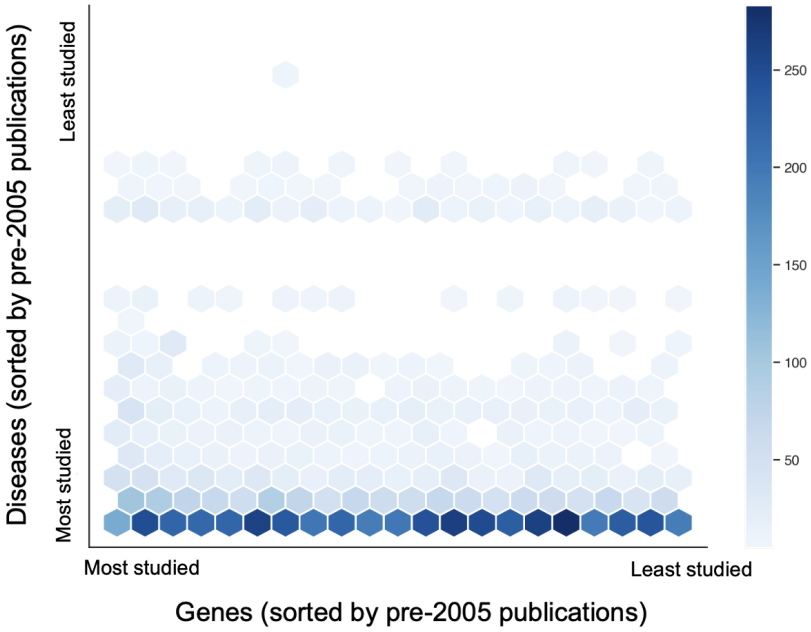
Note: Panel (a) shows the title of the candidate gene study by Qu et al. (2007) published in 2007. Panel (b) shows the title of the GWAS by Hakonarson et al. (2007) published in 2007. Both studies searched for the genetic roots of type 1 diabetes and were conducted by the same principal investigator (PI) in the same year. More detail on these studies is provided in Appendix B. Panel (c) depicts how GWAS introduces new discoveries in the combinatorial landscape of all possible gene-disease combinations. Each gene-disease pair has a specific scientific value, ideally captured by the elevation at that location.

Figure 2: Conditional on the choice of disease, GWAS introduce new gene-disease associations spanning a larger portion of the genetic landscape relative to candidate gene studies.

(a) *Gene-disease associations discovered by candidate gene studies*



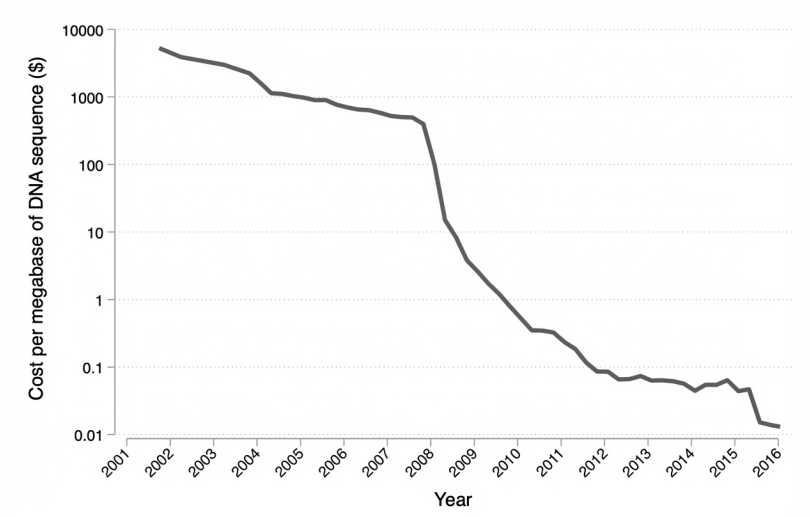
(b) *Gene-disease associations discovered by GWAS*



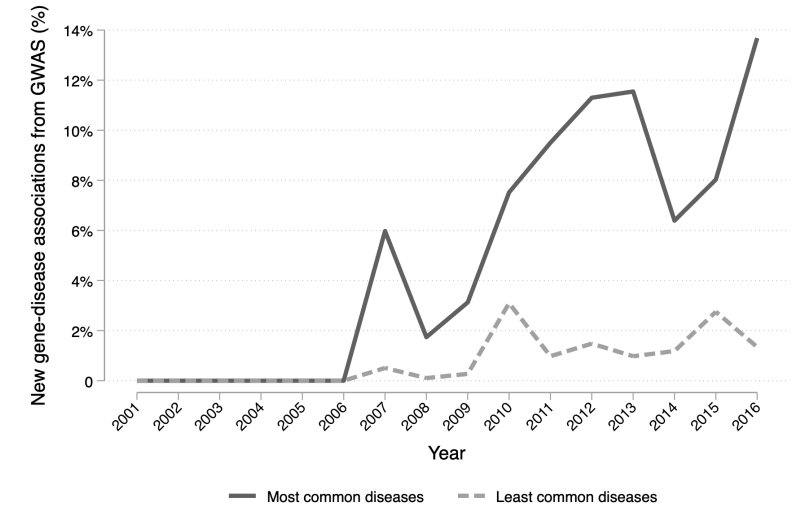
Note: The figure shows the breadth of discovery for different search strategies. Panel (a) shows a heatmap of gene-disease associations introduced after 2005 by candidate gene studies. Panel (b) shows a heatmap of gene-disease associations introduced after 2005 by genome-wide association studies. Both panels show 14,136 genes on the X axis, sorted from most to least studied in the pre-GWAS era, and 9,863 narrowly defined disease categories on the Y axis, sorted from most to least studied in the pre-GWAS era. Darker blue bins correspond to a higher number of associations involving those genes.

Figure 3: The reduction in the cost of DNA sequencing led to the diffusion of GWAS, especially for the most common human diseases.

(a) *Cost of DNA sequencing*



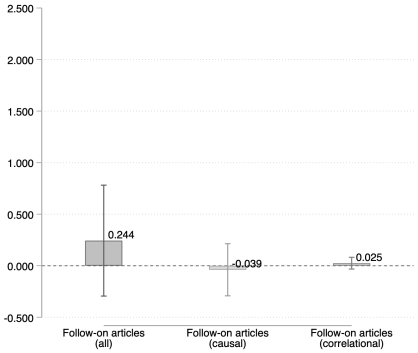
(b) *Share of new gene-disease associations discovered by GWAS*



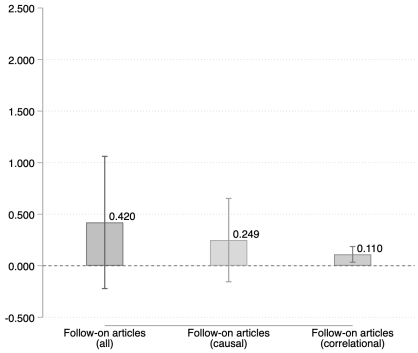
Note: The figure illustrates the intuition behind the shift-share instrumental variable (IV) strategy. Panel (a) shows the average cost of sequencing one megabase (i.e., one million bases) of DNA over time, which serves as a proxy for the broader decline in the cost of genetic data. Panel (b) plots the share of new gene-disease associations (GDAs) recorded in DisGeNET that were discovered by GWAS over time. The figure is shown separately for diseases in the top and bottom quartiles of prevalence in the pre-GWAS era. The figure shows that while declining genotyping costs increased the use of GWAS overall, the increase was larger for the most common diseases, due to ease in assembling study samples and the greater availability of research funding. More details are in Appendix F.

Figure 4: Mechanism analysis: GWAS discoveries attract more follow-on research on causal mechanisms when they are anomalous relative to prior genetic knowledge.

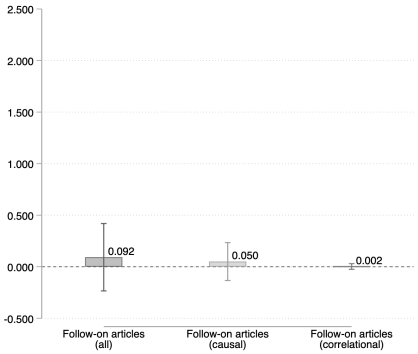
(a) Low-anomaly discoveries, disease areas with limited prior knowledge



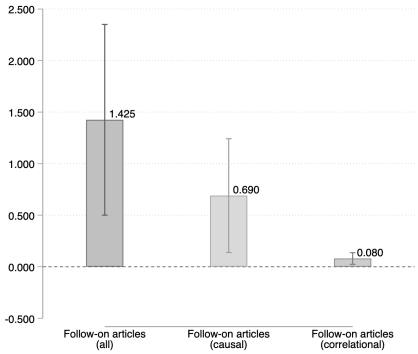
(b) Low-anomaly discoveries, disease areas with extensive prior knowledge



(c) High-anomaly discoveries, disease areas with limited prior knowledge



(d) High-anomaly discoveries, disease areas with extensive prior knowledge

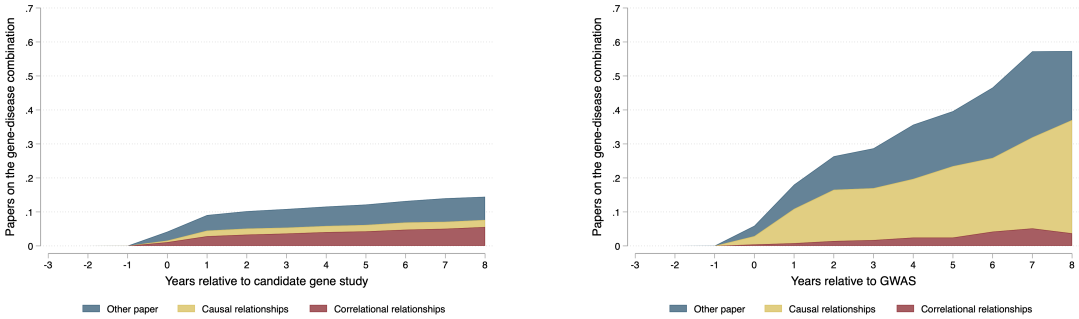


Note: This figure examines how the effect of GWAS on follow-on research varies with the anomalousness of a discovery and the extent of prior genetic knowledge in the focal disease area. Prior genetic knowledge is proxied by the number of gene-disease associations for the disease documented in the pre-GWAS period. Both variables are split at their respective medians, yielding four subsamples: low-anomaly discoveries in diseases with limited (a) or extensive (b) prior genetic knowledge, and high-anomaly discoveries in diseases with limited (c) or extensive (d) prior genetic knowledge. Each panel replicates the cross-sectional specification in Table 3, Panel (b), within the corresponding subsample: $(Follow\text{-}on\ articles)_{i,j} = \alpha + \beta\, GWAS_{i,j} + \gamma\, Disease_j + \omega\, Scientist_k + \delta\, Year_{i,j} + \epsilon_{i,j}$. The charts plot values of β for different dependent variables.

Figure 5: Relative to candidate gene studies, GWAS introduce new gene-disease associations that attract more follow-on research investigating causal mechanisms.

(a) Follow-on articles on gene-disease associations discovered by candidate gene studies

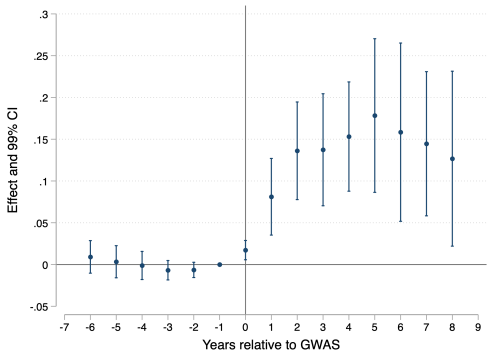
(b) Follow-on articles on gene-disease associations discovered by GWAS



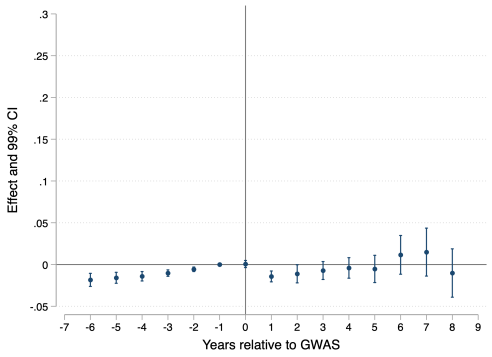
Note: Panel (a) shows the average yearly number of papers exploring gene-disease associations first discovered by a candidate gene study. Panel (b) shows the average yearly number of papers exploring gene-disease associations first discovered by a GWAS. Note that follow-on papers are those directly studying the same gene-disease pair, regardless of whether they cite the study that introduced it. In both figures, the count of papers is split out by whether the follow-on papers explore causal or correlational relationships between the gene and the disease.

Figure 6: Event study regressions show that GWAS discoveries sustain a durable increase in follow-on research investigating causal mechanisms.

(a) Increase in articles investigating causal relationships about the gene-disease combination



(b) Increase in articles investigating correlational relationships about the gene-disease combination



Note: This figure shows the event study version of the difference-in-differences regressions evaluating the increase in follow-on publications for new gene-disease associations discovered by GWAS relative to new gene-disease associations introduced by candidate gene studies. Each panel shows the event study coefficients estimated from the following specification: $Papers_{i,j,t} = \alpha + \sum_z \gamma_z Post_z \times Publication_{i,j} \times 1(z) + \sum_z \beta_z Post_z \times Publication \times GWAS_{i,j} \times 1(z) + \lambda GD_{i,j} + \delta_t \times Gene_i + \omega_t \times Disease_j + \epsilon_{i,j,t}$. The charts plot values of the triple-difference coefficient β_z for different lags z before and after the gene-disease association is first reported by a GWAS with 99% confidence intervals. Standard errors are clustered at the gene and disease level. The dependent variables are the yearly count of papers investigating causal relationships about the gene-disease pair (panel (a)) and the yearly count of papers investigating correlational relationships about the gene-disease pair (panel (b)).

Table 1: Descriptive statistics.

	Panel A: paper-level descriptives (cross sectional)											
	Candidate gene studies						GWAS					
	mean	median	st d	min	max	N	mean	median	st d	min	max	N
Associations per paper	2.588	2	5.158	1	916	141,395	6.934	3	14.089	1	244	1,250
Genes per paper	1.531	1	2.378	1	641	141,395	4.508	2	7.722	1	88	1,250
Year of publication	2011.07	2011	3.394	2005	2016	141,395	2012.28	2012	2.527	2005	2016	1,250

	Panel B: gene-disease level descriptives (cross sectional)											
	Candidate gene studies						GWAS					
	mean	median	st d	min	max	N	mean	median	st d	min	max	N
I(GDA with never associated gene>0)	0.121	0	0.326	0	1	360,647	0.404	0	0.491	0	1	8,655
I(GDA with recently discovered gene>0)	0.098	0	0.298	0	1	360,647	0.256	0	0.437	0	1	8,655
DisGeNET Score of the GDA	0.056	0.01	0.100	0.01	1	360,647	0.069	0.01	0.122	0.01	1	8,655
I(GDA in bottom 50% DisGeNET Score>0)	0.500	0	0.500	0	1	360,647	0.576	1	0.494	0	1	8,655
I(GDA in top 10% DisGeNET Score>0)	0.094	0	0.291	0	1	360,647	0.150	0	0.357	0	1	8,655
Follow-on articles on the GDA (all)	0.570	0	3.199	0	447	360,647	1.054	0	8.904	0	459	8,655
Follow-on articles on the GDA (causal)	0.198	0	1.469	0	276	360,647	0.513	0	5.020	0	271	8,655
Follow-on articles on the GDA (correlational)	0.129	0	1.047	0	235	360,647	0.021	0	0.378	0	23	8,655
Anomaly Index	0.025	0	0.135	0	1	357,840	0.105	0.002	0.291	0	1	8,617
Year of discovery	2011.04	2011	3.385	2005	2016	360,647	2012.64	2013	2.614	2005	2016	8,655

Note: Associations per paper= number of new gene-disease associations (GDAs) introduced by the focal article; *Genes per paper*= number of genes associated with a disease by the focal article; *Year of publication*= year in which the focal article is published. *I(GDA with never associated gene>0)*: 0/1 = 1 if the GDA includes a gene never associated with a disease before 2005; *I(GDA with recently discovered gene>0)*: 0/1 = 1 if the GDA includes a gene discovered after the year 2000 (i.e., after the Human Genome Project); *DisGeNET Score of the GDA*= average DisGeNET Score of GDAs' scientific quality; *I(GDA in bottom 50% DisGeNET Score>0)*: 0/1 = 1 if the GDA falls below the median DisGeNET Score; *I(GDA in top 10% DisGeNET Score>0)*: 0/1 = 1 if the GDA falls above the top 90th percentile of DisGeNET Score; *Follow-on articles on the GDA (all)*: count of all subsequent studies investigating the GDA; *Follow-on articles on the GDA (causal)*: count of subsequent studies investigating causal relationships between the gene and the disease; *Follow-on articles on the GDA (correlational)*: count of subsequent studies investigating correlational relationships between the gene and the disease; *Anomaly Index*= a synthetic measure of how unlikely a new GDA is relative to prior genetic knowledge; *Year of discovery*= year in which the article introducing the GDA is published. The table reports only data that are effectively used in the empirical estimates, i.e., excluding observations that are dropped by the inclusion of fixed effects.

Table 2: GWAS are more likely to discover gene-disease associations involving less-studied genes relative to candidate gene studies.

	I(GDA with never associated gene>0)		I(GDA with recently discovered gene>0)	
	(1)	(2)	(3)	(4)
GWAS (0/1)	0.261***	0.188***	0.144***	0.108***
	(0.0129)	(0.0152)	(0.0110)	(0.0131)
Disease FE	YES	YES	YES	YES
Year of discovery FE	YES	YES	YES	YES
Principal investigator FE	NO	YES	NO	YES
Observations	369,302	348,921	369,302	348,921
Number of diseases	9,863	9,504	9,863	9,504

Note: *, **,*** denote significance at the 5%, 1%, and 0.1% level, respectively. Cross-sectional regressions at the gene-disease association (GDA) level. Std. err. clustered at the gene and disease level. *GWAS*: 0/1 = 1 for new gene-disease associations introduced by a GWAS. See Appendix Table G.1 for the corresponding results using continuous versions of the dependent variables.

Table 3: GWAS discover gene-disease associations of higher scientific quality and that receive more follow-on research relative to candidate gene studies.

Panel A: Effects of GWAS on the quality of discovery

	DisGeNET Score of the GDA		I(GDA in bottom 50% DisGeNET Score>0)		I(GDA in top 10% DisGeNET Score>0)	
	(1)	(2)	(3)	(4)	(5)	(6)
GWAS (0/1)	0.0117*	0.0111*	0.0625***	0.0369*	0.0411**	0.0355*
	(0.00454)	(0.00476)	(0.0148)	(0.0170)	(0.0140)	(0.0149)
Disease FE	YES	YES	YES	YES	YES	YES
Year of discovery FE	YES	YES	YES	YES	YES	YES
Principal investigator FE	NO	YES	NO	YES	NO	YES
Observations	369,302	348,921	369,302	348,921	369,302	348,921
Number of diseases	9,863	9,504	9,863	9,504	9,863	9,504

Panel B: Effects of GWAS on follow-on science

	Follow-on articles on the GDA (all)		Follow-on articles on the GDA (causal)		Follow-on articles on the GDA (correlational)	
	(1)	(2)	(3)	(4)	(5)	(6)
GWAS (0/1)	0.816***	0.543**	0.450***	0.288**	-0.00253	0.0611***
	(0.146)	(0.188)	(0.0748)	(0.109)	(0.0114)	(0.0161)
Disease FE	YES	YES	YES	YES	YES	YES
Year of discovery FE	YES	YES	YES	YES	YES	YES
Principal investigator FE	NO	YES	NO	YES	NO	YES
Observations	369,302	348,921	369,302	348,921	369,302	348,921
Number of diseases	9,863	9,504	9,863	9,504	9,863	9,504

Note: *, **,*** denote significance at the 5%, 1%, and 0.1% level, respectively. Cross-sectional regressions at the gene-disease association (GDA) level. Std. err. clustered at the gene and disease level. *GWAS*: 0/1 = 1 for new gene-disease associations introduced by a GWAS.

Table 4: Instrumental variable estimates confirm that GWAS introduce gene-disease associations with higher scientific quality and that generate more follow-on research relative to candidate gene studies.

Panel A: Causal effects of GWAS on the quality of discovery

	First Stage	Second Stage (2SLS)		
	GWAS (0/1)	DisGeNET Score of the GDA	I(GDA in bottom 50% DisGeNET Score>0)	I(GDA in top 10% DisGeNET Score>0)
	(1)	(2)	(3)	(4)
IV: Genotyping cost shift-share	-0.0104***			
	(0.00235)			
GWAS (0/1)		0.155***	0.0374	0.429**
		(0.0186)	(0.0280)	(0.154)
Kleibergen-Paap F-statistic	19.5			
Disease FE	YES	YES	YES	YES
Year of discovery FE	YES	YES	YES	YES
Observations	88,581	88,581	88,581	88,581
Number of diseases	1,724	1,724	1,724	1,724

Panel B: Causal effects of GWAS on follow-on science

	First Stage	Second Stage (2SLS)		
	GWAS (0/1)	Follow-on articles on the GDA (all)	Follow-on articles on the GDA (causal)	Follow-on articles on the GDA (correlational)
	(1)	(2)	(3)	(4)
IV: Genotyping cost shift-share	-0.0104***			
	(0.00235)			
GWAS (0/1)		3.131***	1.736***	2.448
		(0.652)	(0.422)	(2.000)
Kleibergen-Paap F-statistic	19.5			
Disease FE	YES	YES	YES	YES
Year of discovery FE	YES	YES	YES	YES
Observations	88,581	88,581	88,581	88,581
Number of diseases	1,724	1,724	1,724	1,724

Note: *, **,*** denote significance at the 5%, 1%, and 0.1% level, respectively. Cross-sectional regressions at the gene-disease association (GDA) level. Std. err. clustered at the gene and disease class level. All models include dummies controlling for disease and year fixed effects. Note that the number of observations is smaller because some diseases lack prevalence data for the pre-GWAS period. The first stage regresses the *GWAS* indicator on the instrument *Genotyping cost shift-share*. The second stage uses the fitted values of *GWAS* as the regressor.

Table 5: Difference-in-differences estimates show that GWAS discoveries attract follow-on higher-impact research published in more prestigious journals.

	Articles weighted by scientific citations (1)	Articles weighted by clinical citations (2)	Articles weighted by journal SJR (3)
Post × Publication (0/1)	4.834***	0.0843***	5.192***
	(0.280)	(0.00697)	(0.120)
... × GWAS (0/1)	20.38***	0.375***	13.33***
	(2.969)	(0.0676)	(1.919)
Gene-disease FE	YES	YES	YES
Disease × year FE	YES	YES	YES
Gene × year FE	YES	YES	YES
Observations	4,782,071	4,782,071	4,782,071
Number of diseases	9,853	9,853	9,853

Note: *, **,*** denote significance at the 5%, 1%, and 0.1% level, respectively. Difference-in-differences panel regressions at the gene-disease-year level. Std. err. clustered at the gene and disease level. *Post* × *Publication*: 0/1 = 1 in all years after a gene-disease pair is first reported in a study; *GWAS*: 0/1 = 1 for gene-disease associations introduced by a GWAS.

From Big Data to "Big Theory"?
Evidence from Genome-Wide Association Studies
Online Appendix

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A Additional Details on Genomic Research

A.1 Scientific Background

Genomics is the branch of biological science focused on the study of genomes—that is, the complete set of an organism's genes. Genes are sequences of DNA bases that encode the "instructions" for synthesizing gene products, most notably proteins. Genes play a fundamental role in the functioning of the human body, but their sequences can sometimes acquire mutations. When this happens, genes may alter their behavior, sometimes leading to emergence of health conditions. At the same time, the role of genes in the etiology of disease creates opportunities for therapeutic intervention: genes causally linked to a condition can often serve as drug targets (Nelson et al., 2015). When a drug molecule binds to its target, it can modify the gene's function, potentially improving the condition. As a result, understanding the genetic roots of disease has direct implications for the design of pharmaceutical drugs.

Genetic research has historically been very effective at identifying genes that are individually responsible for specific disease conditions. These rare diseases are known as Mendelian disorders, usually visible from birth and traceable through family history. Understanding the causes of Mendelian disorders was one of the earliest successes of genetic research (Bush and Moore, 2012). Take, for example, the case of cystic fibrosis. This rare disorder is caused by DNA mutations that tend to cluster in the Cystic Fibrosis Transmembrane Conductance Regulator (CFTR) gene. This discovery was made by genotyping families affected by cystic fibrosis. Despite small sample sizes, due to the rarity of the disease, the strong effect and localization of the responsible mutations in a single gene made it possible to unambiguously identify the gene as causally linked to the condition.

However, Mendelian diseases are rare because their severity renders them susceptible to negative evolutionary selection. Much more common are polygenic diseases, which are caused by mutations in multiple genes. In these cases, any mutation may increase disease risk without being either necessary or sufficient, meaning that it usually accounts for only a small fraction of the condition's heritability. Although complex disorders also cluster within families, they do not follow predictable inheritance patterns, as they are shaped by intricate interactions between genetic and environmental factors. For this reason, family-based studies have proven less effective when applied to common polygenic diseases.

A.2 A Scientific Primer on GWAS

Towards the end of the last century, researchers increasingly embraced the idea that common disorders are likely influenced by genetic mutations that are also common in the population (Reich and Lander, 2001).

Rather than searching for individual genes with strong effects, the field has shifted toward studying common variants that individually have only a small impact on disease risk (Bush and Moore, 2012). But what exactly is a variant? At the most basic level, two genomes differ at a specific genetic location if they contain different single nucleotides (adenine, thymine, cytosine, or guanine) at that position. When such a difference occurs in at least 1% of the population, it is referred to as a single-nucleotide polymorphism (SNP). Connecting SNPs with diseases is based on the idea that a causative variant should appear more frequently in individuals with the disease than in those without it. In practice, researchers search for statistical associations between variants and diseases in large population samples.

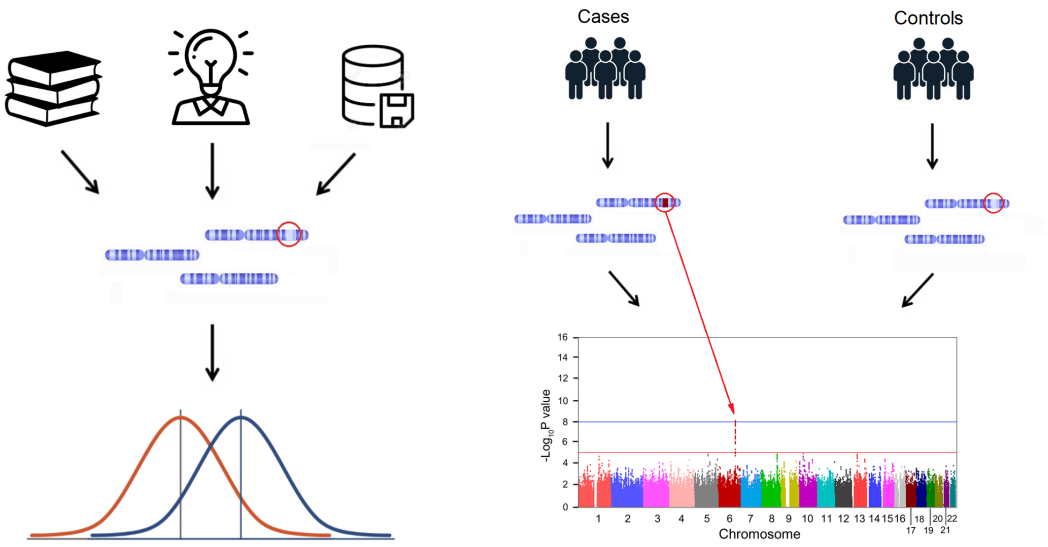
Building on this logic, candidate gene studies are a theory-driven approach used in genetic research to identify associations between specific genes and diseases. Panel (a) of Figure A.1 shows a stylized representation. Researchers begin by selecting one or more genes based on prior biological knowledge, typically genes believed to be involved in the physiological pathways relevant to the disease under investigation. The process involves genotyping individuals in a case-control design to detect whether genetic variants in the candidate genes occur more frequently in individuals with the disease than in those without it. Statistical tests are then used to determine whether these variants are significantly associated with the presence or severity of the disease. Note that this approach genotypes individuals only at the specific locations hypothesized as important. Candidate gene studies are thus limited by their reliance on existing theory, which can bias discovery toward well-known genes and overlook novel or unexpected genetic contributors.

In contrast, genome-wide association studies (GWAS) are hypothesis-free methods for identifying gene-disease associations (Visscher et al., 2017). Panel (b) of Figure A.1 shows a representation. GWAS are also case-control studies: researchers collect DNA from patients with the disease under study and individuals without it. However, in this case, researchers rely on high-throughput microarrays to collect data on millions of genetic variants across the entire genome of the individual. Compared to candidate gene studies, GWAS typically require much larger sample sizes to achieve statistical significance, given the large number of hypotheses tested, thereby increasing their costs. It is important to note that array-based GWAS do not sequence DNA base by base, since they only detect the presence or absence of specific SNPs. While microarrays can genotype millions of SNPs, they still cover less than 0.1% of the genome. However, because the co-occurrence of nearby genetic variants is not random (a phenomenon known as linkage), researchers can use reference genomes to efficiently infer the characteristics of the broader genome from the smaller subset of SNPs directly genotyped (Bush and Moore, 2012; Uffelmann et al., 2021).[1]

[1]GWAS collect data only on selected genetic loci whose variation can be extrapolated to represent their surrounding genetic regions. As a result, researchers can efficiently infer the broader structure of the genome by sampling only a tiny fraction of DNA bases

Figure A.1: Stylized comparison of two types of genetic studies.

(a) *Schema of a typical candidate gene study* (b) *Schema of a typical GWAS*



Note: Panel (a) shows a schema of how a candidate gene study unfolds. First, researchers select the disease of interest and decide which gene(s) to investigate based on previous literature, existing evidence, or their own intuition. Then, the genome of people with and without the condition is genotyped *at that genetic location* in search of differences. Finally, statistical methods are used to test the association between mutations in the targeted genes and the disease of interest. Panel (b) shows a schema of how a GWAS unfolds. First, researchers select the disease of interest, but not the genes. Then, *the entire genome* of people with and without the condition is genotyped in search of genetic differences. Finally, statistical methods are used to test the association between mutations in each gene and the disease of interest. The results are often graphically represented as a "Manhattan plot," which shows the p-value of multiple statistical tests comparing DNAs between the case and control groups. The y-axis reports $-log_{10}(P\ value)$, so higher values correspond to stronger associations.

GWAS have led to important scientific discoveries, but have also come under scrutiny for several limitations. First, genome-wide scans detect common variants that only serve as markers for broader genomic regions likely to contain causal mutations. However, GWAS cannot pinpoint the exact causal SNPs with certainty, so additional analyses or follow-up studies are typically required to narrow the region of association. Second, even when the relevant mutation is identified, GWAS cannot explain the biological mechanism underlying its role in human health, which is essential for drug discovery (Nelson et al., 2015). Third, most complex diseases are influenced by many genes, so the contribution of any single variant is usually small. This limits the therapeutic value of GWAS findings, as variants with small effects may not be actionable drug targets (Goldstein, 2009). Furthermore, many of such discoveries later prove to be false positives (Tranchero, 2026). Finally, critics have noted that GWAS tend to overlook more complex genetic architectures, since they are primarily designed to test pairwise gene-disease associations (Boyle et al., 2017). As a result, their value

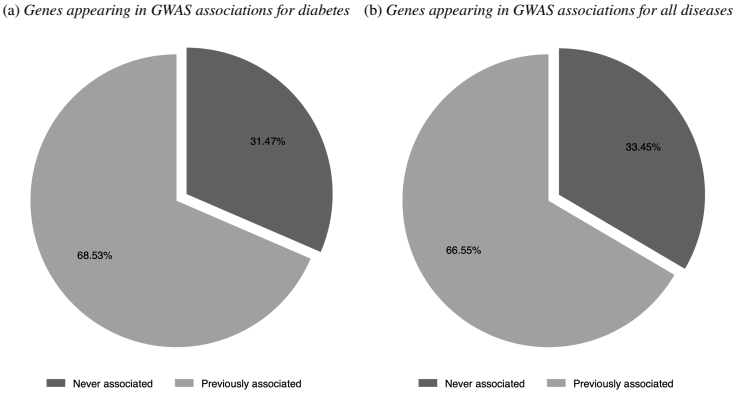
(Visscher et al., 2017). This approach differs from more recent whole-genome sequencing (WGS), which reads every DNA base, and has diffused in the past few years.

remains debated (Callaway, 2017).

A.3 GWAS as "Unbiased" Data-Driven Search

A central feature of GWAS is that they are designed to screen genetic variation across the entire genome, without restricting attention to genes selected based on prior theory. Major reviews emphasize that *"GWASs are unbiased with respect to prior biological knowledge (or prior beliefs) and with respect to genome location"* (Visscher et al., 2012). This design choice is often summarized by describing GWAS as a *"hypothesis-free method"* for identifying associations between genes and diseases.[2] While this enhances the potential for discovery when biological understanding is incomplete (Pearson and Manolio, 2008), the design of GWAS also limits investigators' ability to steer the genetic search toward particular gene-disease links, since the same association-testing procedure is applied to the entire genome. Reporting norms limit selective disclosure, because GWAS publications conventionally present the full set of statistically significant associations emerging from the scan rather than a hand-picked subset of "novel" genes.

Figure A.2: Rediscovery of known genes in GWAS-reported associations.



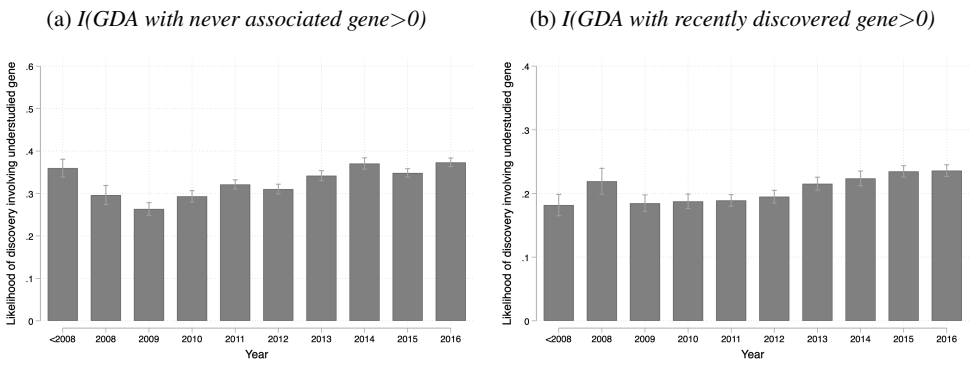
Note: Each panel reports the share of genes that appear in GWAS that were already linked to the corresponding disease prior to 2005 ("known") versus genes that were not previously linked ("novel"). Panel (a) restricts to GWAS on diabetes, while panel (b) aggregates across all diseases in the sample.

Several falsification tests support interpreting GWAS as an *unbiased* data-driven search procedure. First, because GWAS implement a genome-wide screen, studies should not only surface novel associations but also re-identify previously established ones. This is visible in the motivating example of Hakonarson et al. (2007): Figure B.1 shows that alongside a novel association involving *KIAA0350*, the GWAS also re-discovers known associations at four genes (*PTPN22*, *INS*, *COL1A2*, *LPHN2*). The same pattern appears across all studies in my sample, for which 66.5% of genes reported as GWAS discoveries were already known prior to 2005

[2]Source: https://www.ebi.ac.uk/gwas-catalogue/what-are-genome-wide-association-studies/.

(Figure A.2). Second, I examine whether the observable characteristics of GWAS discoveries display time trends, which would be inconsistent with a stable genome-wide screening procedure. Figure A.3 shows that GWAS yield similar kinds of associations across years.

Figure A.3: The novelty yield of GWAS is stable over time.



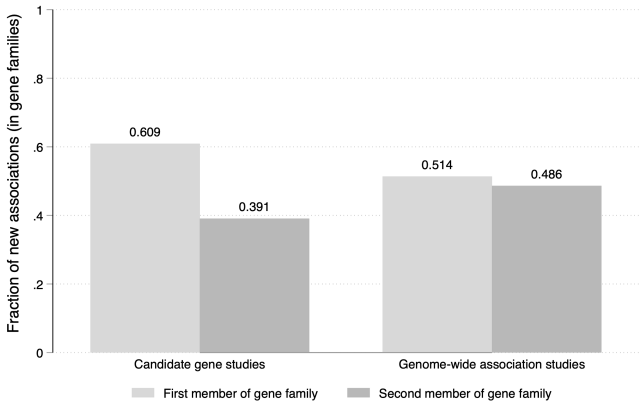
Note: The figure plots the annual share of new gene-disease associations introduced by GWAS involving (a) genes never associated with any disease before 2005 and (b) genes discovered after 2000. Years before 2008 are pooled.

Third, I examine how GWAS and candidate gene studies distribute their discoveries across members of the same gene family (Daugherty et al., 2012). Genes within a family share related biological functions and are typically numbered in order of discovery (e.g., BRCA1 and BRCA2, discovered in 1994 and 1995, respectively), thus differing in the extent to which they have been studied. In the DisGeNET data, the first-numbered member of a family received, on average, 49% more publications than the second. Yet, associations involving the second member have slightly higher average DisGeNET Scores (0.063 vs. 0.059) and are more likely to fall in the top 10% (11.2% vs. 9.8%).

Figure A.4 shows that candidate gene studies disproportionately introduce associations involving the first-discovered family member, whereas this pattern largely disappears for GWAS. In regressions controlling for disease and year fixed effects, GWAS are 8.7 percentage points more likely to introduce associations involving the second family member ($p < 0.001$). This pattern is consistent with a genome-wide screening that reduces the concentration of discovery on more familiar genes.

Finally, I test whether novelty is rewarded in the GWAS publication process. If researchers had strong incentives to pursue novel findings, then GWAS reporting more novel or more counterintuitive gene-disease associations should systematically appear in more prestigious outlets. Figure A.5 shows that the novelty or degree of anomaly of GWAS discoveries is not associated with better publication outcomes. The absence of a novelty premium suggests that GWAS researchers face limited incentives to strategically emphasize novel associations in what they report, reinforcing the interpretation that the novelty content of GWAS findings is

Figure A.4: Candidate gene studies concentrate more heavily on first-numbered members of gene families than GWAS.

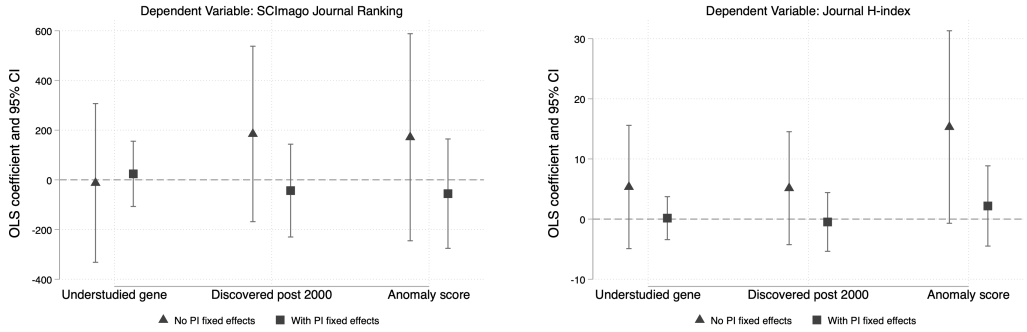


Note: The figure shows the share of new gene-disease associations involving the first versus the second member of a gene family, separately by the method used in the paper introducing the association.

largely a product of genome-wide screening.

Figure A.5: The novelty of GWAS discoveries is not related to better publication outcomes.

(a) *Journal prestige (SCImago rank)* (b) *Journal impact (journal h-index)*



Note: Each panel relates the novelty or anomaly of GWAS-reported associations to publication outcomes. Panel (a) measures journal prestige using the SCImago Journal Rank, and panel (b) measures journal citation impact using the journal h-index.

B Detailed Case Study: Qu et al. (2007) vs. Hakonarson et al. (2007)

B.1 Genetic Research on type 1 diabetes

It is estimated that around 40 million Americans have diabetes (source: https://www.cdc.gov/diabetes). The two main forms of diabetes are type 1 and type 2, with type 1 accounting for approximately 5–10% of cases. Type 1 diabetes is a chronic condition that typically begins in childhood. More specifically, it is an autoimmune disease caused by the immune system's destruction of pancreatic β-cells (DiMeglio et al., 2018). As a result, the pancreas is unable to produce sufficient insulin, the hormone essential for enabling sugar to enter cells, produce energy, and regulate blood glucose levels. To date, there is no known way to prevent type 1 diabetes, and lifelong insulin therapy remains necessary for patient survival.

Genetics plays an important role in the onset of type 1 diabetes: children with a parent affected by the condition face a risk of 1–9% of developing it themselves (DiMeglio et al., 2018). Early candidate gene studies identified several genetic determinants, primarily within a tightly linked group of genes known as the major histocompatibility complex (MHC). MHC genes encode cell surface proteins that are essential for initiating and directing the immune response. When these genes malfunction, they can trigger autoimmune activity, such as the immune system attacking the body's own β-cells in type 1 diabetes. However, MHC genes account for only slightly more than half of the genetic risk associated with the disease, suggesting that other, still unidentified, genes are involved. A systematic effort to detect these remaining genes could reveal alternative therapeutic targets and shed light on the deeper causes of type 1 diabetes.

B.2 A Tale of Two Studies

In June 2007, Qu et al. (2007) published a candidate gene study based on a sample of 947 family trios with type 1 diabetes (i.e., one affected child and both parents). Using a targeted sequencing assay, the researchers tested the association between type 1 diabetes and a specific gene: interferon regulatory factor 5 (IRF5). The choice of IRF5 was informed by prior findings linking it to autoimmune diseases such as systemic lupus erythematosus. Based on these similarities, the authors hypothesized that IRF5 might also play a role in type 1 diabetes, another autoimmune condition with overlapping features. As they put it, the *"[...] association of IRF5 with other autoimmune diseases, such as T1D, has a high prior probability"* based on existing evidence and genetic understanding. However, despite being grounded in theoretical reasoning and prior knowledge, the hypothesis turned out to be incorrect. The results showed no significant association between mutations in IRF5 and type 1 diabetes risk, leading Qu et al. (2007) to conclude that their proposed gene-disease combination had limited therapeutic potential.

Figure B.1: Main results from the GWAS analysis of Hakonarson et al. (2007).

(a) *Results from main analysis*

Case-control cohort				
Chr.	SNP	OR (95% CI)	*P*-value	Locus
1	rs2476601	1.80 (1.44, 2.24)	1.32×10^{-7}	*PTPN22*
11	rs1004446	0.62 (0.53, 0.73)	4.38×10^{-9}	*INS*
16	rs2903692	0.65 (0.56, 0.76)	4.77×10^{-8}	*KIAA0350*
11	rs6356	1.52 (1.31, 1.76)	1.78×10^{-8}	*INS*
16	rs725613	0.67 (0.58, 0.78)	3.24×10^{-7}	*KIAA0350*
7	rs10255021	0.58 (0.44, 0.77)	1.16×10^{-4}	*COL1A2*
11	rs10770141	0.65 (0.56, 0.76)	7.20×10^{-8}	*INS*
1	rs672797	1.54 (1.29, 1.85)	2.67×10^{-6}	*LPHN2*
16	rs17673553	0.66 (0.55, 0.78)	1.30×10^{-6}	*KIAA0350*
11	rs7111341	0.63 (0.53, 0.76)	3.77×10^{-7}	*INS*
11	rs10743152	0.67 (0.57, 0.78)	4.73×10^{-7}	*INS*

(b) *Text excerpt on KIAA0350 gene*

This locus resides in a 233-kb block of LD that contains only *KIAA0350* and no other genes, making this gene a prime candidate for harbouring the causative variant. *KIAA0350* encodes a protein of unknown function and its genomic location is next to the suppressor of cytokine signalling 1 (*SOCS1*) gene. The almost exclusive expression specificity of *KIAA0350* in immune cells (http://symatlas.gnf.org/SymAtlas), including dendritic cells, B lymphocytes and natural killer (NK) cells, all of which are pivotal in the pathogenesis of T1D[27,28], indicates that the variant probably contributes to the disease by modulating immunity.

Note: Panel (a) shows an excerpt from Table 1 of Hakonarson et al. (2007). The genetic location of the SNPs associated with type 1 diabetes is shown in the rightmost column. Highlighted in red are the three genetic variants located in the KIAA0350 gene. Panel (b) shows the passage of Hakonarson et al. (2007) describing the inferred role of KIAA0350. T1D stands for type 1 diabetes.

Two months later, in August 2007, Hakonarson et al. (2007) published a GWAS based on a study population of 563 patients with type 1 diabetes and 1,146 controls.[3] The analysis was conducted using a microarray capable of genotyping 550,000 single nucleotide polymorphisms (SNPs) across the whole genome. The study identified several SNPs associated with type 1 diabetes, as shown in Figure B.1. Some of the significant SNPs were found in genes already known to be related to diabetes (for example, the insulin gene INS), but three were located in KIAA0350. It is now understood that this gene helps regulate β-cell function and thus plays a role in preventing diabetes. However, KIAA0350 was among the least studied human genes at the time, and its function was largely unknown (Soleimanpour et al., 2014). The association between KIAA0350 and type 1 diabetes proved robust and has since been investigated in multiple follow-up studies, contributing to a deeper genetic understanding of the disease (Gingerich et al., 2020).

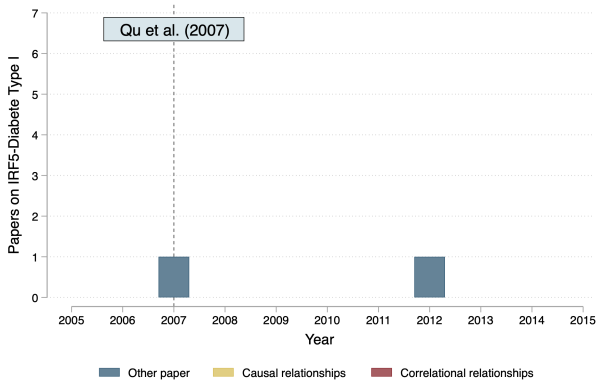
B.3 The Effect of GWAS on Follow-on Research

The studies by Hakonarson et al. (2007) and Qu et al. (2007) offer a clean comparison because they were conducted in the same year, on the same disease, and by the same principal investigator (PI). Figure B.2 summarizes the amount of follow-on research each gene-disease combination received, based on DisGeNET. The candidate-gene study of Qu et al. (2007) generated little interest (Panel (a)), maybe not surprisingly given its weak findings. In the IRF5 case, prior knowledge guided attention to a familiar and already intensively studied gene, so the study's main contribution was an assessment of an existing theory-driven conjecture. Instead, the GWAS association with KIAA0350 (now renamed as CLEC16A) triggered substantial follow-on research, mostly aimed at understanding if and how it is causally tied to diabetes.

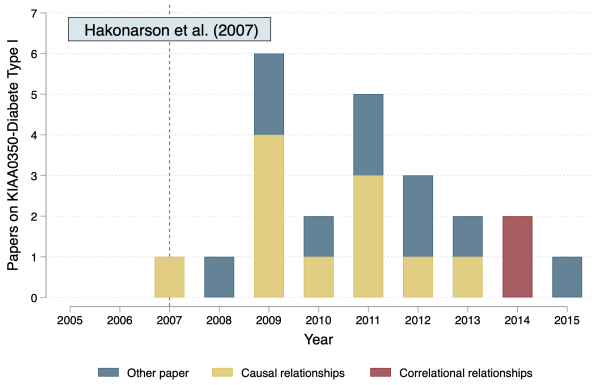
[3]The study also replicated its main analysis in a separate sample of 483 nuclear family trios, leveraging genetic differences between affected children and their parents.

Figure B.2: Follow-on papers investigating the association between type 1 diabetes and the IRF5 and KIAA0350 genes, respectively.

(a) *Papers on IRF5-T1D after Qu et al. (2007)*



(b) *Papers on KIAA0350-T1D after Hakonarson et al. (2007)*



Note: Panel (a) shows the yearly count of publications exploring the relationship between IRF5 and type 1 diabetes following the candidate gene study by Qu et al. (2007). Panel (b) shows the yearly count of publications exploring the relationship between KIAA0350 and type 1 diabetes following the GWAS by Hakonarson et al. (2007). Data on publications and the gene-diseases studied come from DisGeNET, while information on the type of relationship studied comes from PubTator3. In both panels, the paper introducing the gene-disease combination is excluded.

A qualitative reading of these papers suggests how the GWAS finding influenced subsequent research. A first set of follow-on papers treats the GWAS association as an identification problem and asks exactly which genetic variant in the KIAA0350 locus could be responsible for disease risk. Because the original association spans a broader region, subsequent work concentrates on fine mapping and functional annotation. In so doing, studies building on Hakonarson et al. (2007) aim to specify how mutations in the KIAA0350 region could plausibly increase the risk of type 1 diabetes. In this sense, the anomaly created an impetus to reconcile genotype-level evidence with a biologically interpretable target.

A second set of follow-on papers treats the locus as a mechanistic puzzle and asks what the implicated gene does in disease-relevant tissue, and through what biological pathway. This work moves from association to theory-building by proposing testable hypotheses, identifying molecular interaction pathways, and using experimental methods to establish causality. For example, later work characterizes KIAA0350 as an endosomal protein and links it to mitochondrial quality control through mitophagy in pancreatic β-cells, with downstream effects on insulin secretion and glucose homeostasis (Soleimanpour et al., 2014). This exemplifies the core dynamics documented by my results: an anomalous GWAS discovery motivates theorizing by prompting researchers to build explanations that connect a gene to a specific physiological function. The empirical footprint of this process is an increase in studies that make mechanism-oriented claims, such as demonstrating that altering gene function changes disease-relevant phenotypes.

C DisGeNET Data

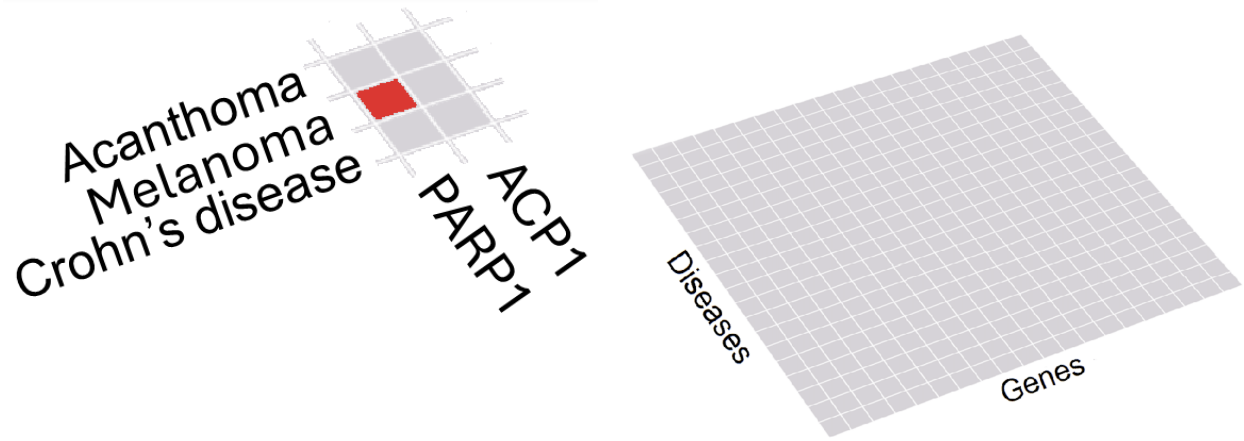
This appendix describes the empirical approach used in the paper and provides additional details on the DisGeNET data.

C.1 The Gene-Disease Search Landscape

One advantage of focusing on a specific search problem, such as identifying the genetic roots of human diseases, is that the search landscape is well-defined. The relevant components in this combinatorial problem are genes and diseases. In principle, any gene could contain mutations that contribute to a given condition. This means the search task can, in first approximation, be reduced to identifying gene–disease pairs $< i, j >$ in which mutations in gene i are causally linked to the emergence of disease j (Panel (a) of Figure C.1). Another benefit of this setting is that all the relevant components of the problem have already been mapped and codified in exhaustive taxonomies. The Human Genome Project cataloged approximately 19,000 protein-coding genes and assigned them unique names and identifiers. Similarly, thousands of human diseases have been classified in systems such as the Unified Medical Language System (UMLS), which provides unique disease identifiers. The Cartesian product of these entities defines the full combinatorial landscape in which search takes place, as illustrated in Panel (b) of Figure C.1.

Figure C.1: Empirically tracing the combinatorial landscape to study search.

(a) *Example of gene-disease combinations* (b) *The gene-disease combinatorial landscape*



Note: Panel (a) shows an example of gene-disease combination. Genes and diseases can be sorted based on similarity or relatedness, such as melanoma and acanthoma, which are both skin tumors. The figure identifies in red the combination of PARP1 and melanoma, which was introduced by a GWAS. Panel (b) shows the idealized combinatorial landscape of all possible gene-disease pairs.

In terms of empirical design, the combinatorial landscape represents the "ground truth," that is, the space of entities over which researchers conduct their search activities. Mapping this knowledge landscape allows

for the empirical study of how actors search across it (Aharonson and Schilling, 2016). Specifically, one can locate research outputs (e.g., scientific publications) on the landscape by extracting the relevant entities (e.g., genes and diseases), thereby characterizing search in spatial terms. This approach offers two main advantages. First, it enables researchers to record the features of each combination relative to its position on the landscape. For example, Figure 2 illustrates how the spatial distribution of new combinations can be used to assess whether a search is "local" or "distant." Second, treating search as a landscape-based activity removes the need for citations to track knowledge evolution. Instead, one can observe changes in follow-on research that directly engages with the same entities before and after a given study. For instance, Figure 5 plots the yearly number of papers focused on a specific gene–disease combination, regardless of whether they cite the original study that introduced it.

This empirical approach extends beyond the specific setting of human genetics. It can be applied more broadly wherever the relevant search landscape can be defined ex ante, independently of the discoveries ultimately made. Advances in data-generating technologies are making this increasingly feasible. From genetic atlases (Kang, 2025; Kao, 2026) to satellite imagery (Nagaraj, 2022), large-scale mapping efforts are rendering previously difficult-to-observe landscapes measurable, enabling researchers to study search processes against a well-defined set of potential alternatives.

C.2 The DisGeNET Score

The main source of data for this paper is DisGeNET (v7.0), a platform integrating information from multiple sources to create a comprehensive repository of scientific findings linking human diseases to their genetic causes (Hermosilla and Lemus, 2019; Piñero et al., 2020). The database compiles gene-disease associations (GDAs) from manually curated datasets, experimental studies on animal models, and literature mining from PubMed. The version used in this paper covers publications through 2016 and includes over 628,000 gene–disease pairs involving 17,549 protein-coding genes and 24,166 diseases. Genes are identified using their NCBI Gene ID (formerly EntrezGene ID), and diseases are coded using UMLS identifiers. The empirical analyses focus on GDAs first introduced between 2005 and 2016. The start date reflects the emergence of GWAS as a research design in 2005, which is necessary to compare genome-wide and targeted approaches during the same time window. Instead, the end date is determined by data availability, since the DisGeNET version I use covers publications through 2016, and later versions are not freely available to researchers.

DisGeNET is designed to help researchers in both academia and industry prioritize promising genetic targets. To support this goal, it provides a synthetic DisGeNET Score for each gene–disease combination. The Score

ranges from 0 to 1, with higher scores indicating associations that are more robust and therapeutically more promising. In the version used in this paper (v7.0), the Score is defined as follows:

$$\text{DisGeNET Score of gene-disease combination} < \text{i, j} > = C_{i,j} + M_{i,j} + I_{i,j} + L_{i,j}$$

The first component $C_{i,j}$ summarizes the evidence from curated sources reporting gene-disease combination $< i, j >$:

$$C_{i,j} = \begin{cases} 0.6 & \text{if } N_{sources_c} > 2 \\ 0.5 & \text{if } N_{sources_c} = 2 \\ 0.3 & \text{if } N_{sources_c} = 1 \\ 0 & \text{otherwise} \end{cases} \tag{C.1}$$

where $N_{sources_c}$ is the number of curated sources supporting a gene-disease association, including CGI, ClinGen, Genomics England, CTD, PsyGeNET, Orphanet, and UniProt.

The second component $M_{i,j}$ summarizes the evidence from experiments using animal models reporting gene-disease combination $< i, j >$:

$$M_{i,j} = \begin{cases} 0.2 & \text{if } N_{sources_m} > 0 \\ 0 & \text{otherwise} \end{cases} \tag{C.2}$$

where $N_{sources_m}$ is the number of sources using the lab rat or lab mouse from RGD, MGD, and CTD.

The third component $I_{i,j}$ summarizes the evidence inferred from experiments on gene-disease combination $< i, j >$:

$$I_{i,j} = \begin{cases} 0.1 & \text{if } N_{sources_i} > 1 \\ 0 & \text{otherwise} \end{cases} \tag{C.3}$$

where $N_{sources_i}$ is the number of sources from HPO and CLINVAR. In the original version of the Score, papers listed in the GWAS Catalog were also included in this count. However, this would generate a mechanical upward bias in the DisGeNET Score of gene-disease pairs introduced by GWAS. Therefore, the current paper excludes those sources from the calculation of the DisGeNET Score, ensuring that my results are not due to a different weighting of GWAS papers.

Finally, the component $L_{i,j}$ summarizes the evidence mined from the literature about gene-disease combi-

nation $< i, j >$:

$$L_{i,j} = \begin{cases} 0.1 & \text{if } N_{publications} > 9 \\ \text{N}_{publications}\text{*0.01} & \text{if } N_{publications} <= 9 \end{cases} \tag{C.4}$$

where $N_{publications}$ is the number of publications supporting a gene-disease association as mined by LHGDN and BEFREE.

C.3 Robustness and Validation of the DisGeNET Score

The DisGeNET Score has strong face validity and has been extensively validated as a metric capturing the scientific strength of gene-disease associations (Piñero et al., 2020). Returning to the earlier example of type 1 diabetes, the IRF5–type 1 diabetes combination has a low DisGeNET Score of 0.03, compared to a much higher score of 0.46 for the KIAA0350–type 1 diabetes pair. These scores align well with the relative scientific and therapeutic importance of the two combinations (Gingerich et al., 2020). Table C.1 shows that publications introducing gene-disease combinations with higher DisGeNET Scores receive substantially more citations. This relationship remains strong when controlling for disease and scientist fixed effects. As an additional validation, Table C.2 compares the DisGeNET Score of each gene-disease combination with downstream innovation and therapeutic outcomes. Combinations with higher Scores are associated with significantly more granted patents, citations from clinical research, and FDA-approved drugs. Importantly, none of these downstream outcomes are used in calculating the Score itself.

Table C.1: The DisGeNET Score of a given gene-disease combination is strongly associated with the citations accrued to the paper first introducing it.

	Citations to the paper introducing the gene-disease pair		
	(1)	(2)	(3)
DisGeNET Score of the gene-disease pair	117.4*** (8.564)	106.1*** (8.802)	65.79*** (5.731)
Year of discovery FE	YES	YES	YES
Disease FE	NO	YES	YES
Principal investigator FE	NO	NO	YES
Observations	374,596	369,305	347,311
Number of diseases	15,155	9,864	9,507
Mean of the DV	45.18	45.18	45.18

Note: *, **,*** denote significance at the 5%, 1%, and 0.1% level, respectively. Cross-sectional regressions at the gene-disease association (GDA) level. Std. err. clustered at the gene and disease level. Column 2 adds disease fixed effects, and Column 3 adds scientist fixed effects. *Citations to the paper introducing the gene-disease pair*: count of scientific citations received by the focal article introducing the new gene-disease combination; data from NIH's iCite is available here: https://icite.od.nih.gov/.

Table C.2: The DisGeNET Score of a given gene-disease combination is strongly associated with more USPTO granted patents, clinical impact, and FDA-approved drugs on the same combination.

	Patents granted (1)	Clinical citations (2)	Drugs approved (3)
DisGeNET Score of the gene-disease pair	1.313*** (0.207)	27.31*** (3.381)	0.0440*** (0.0128)
Disease FE	YES	YES	YES
Year of discovery FE	YES	YES	YES
Observations	369,291	369,291	369,291
Number of diseases	9,862	9,862	9,862
Mean of the DV	0.193	2.128	0.008

Note: *, **,*** denote significance at the 5%, 1%, and 0.1% level, respectively. Cross-sectional regressions at the gene-disease association (GDA) level. Std. err. clustered at the gene and disease level. *Patents granted*: count of USPTO patents granted from 2015 to 2023 that mention in their text a specific gene-disease combination. Data from EBI's SureChEMBL Workshop 2024 is available here: https://ftp.ebi.ac.uk/databases/SureChEMBL/SureChEMBLWorkshop2024/. *Clinical citations*: count of clinical articles up to 2024 citing the paper introducing a specific gene-disease combination. Data from NIH's iCite is available here: https://icite.od.nih.gov/. *Drugs approved*: count of FDA-approved drugs up to 2023 targeting a specific gene-disease combination. Data from DrugCentral 2023 is available here: https://drugcentral.org/.

A potential concern is that the DisGeNET Score incorporates evidence accumulated after the original discovery, which could mechanically link the measured quality of a gene-disease association to the subsequent research it attracts. To assess this concern, I reconstruct the publication history of each association in DisGeNET. Among the 605,691 supporting publication-association pairs in my estimation sample, 36.4% involve publications appearing after the discovery year. However, their contribution to the Score is relatively small. As discussed in Appendix C.2, the literature component increases by 0.01 per supporting publication, up to a maximum of 0.10. I therefore construct an alternative Score excluding the contribution of publications appearing after the discovery year. These publications increase the Score for only 20.6% of associations, and removing their contribution reduces the mean Score by 8.2%. Importantly, the average contribution is slightly smaller for GWAS than for candidate gene discoveries (0.0041 versus 0.0046), indicating that subsequent publications do not mechanically inflate the Score of GWAS discoveries.

Table C.3 examines whether excluding post-discovery literature alters the estimated quality advantage of GWAS. Columns (1)–(2) reproduce the baseline specifications from Table 3, while Columns (3)–(4) use the alternative Score. The GWAS coefficient remains positive and statistically significant, with only a modest reduction in magnitude. Columns (5)–(6) provide an additional check, controlling for the number of supporting publications and independent non-literature sources. The GWAS coefficient remains positive and

statistically significant in both specifications. Taken together, these results indicate that the higher DisGeNET Scores of GWAS discoveries are not driven by the mechanical contribution of subsequent publications.

Table C.3: The higher DisGeNET Scores of GWAS discoveries persist after removing the contribution of post-discovery publications.

	DisGeNET Score		DisGeNET Score excluding post-discovery literature		DisGeNET Score	
	(1)	(2)	(3)	(4)	(5)	(6)
GWAS (0/1)	0.0117*	0.0111*	0.0101*	0.0117*	0.0250***	0.0208***
	(0.00454)	(0.00476)	(0.00443)	(0.00467)	(0.00210)	(0.00221)
Year of discovery FE	YES	YES	YES	YES	YES	YES
Disease FE	YES	YES	YES	YES	YES	YES
Principal investigator FE	NO	YES	NO	YES	NO	YES
Evidence-volume controls	NO	NO	NO	NO	YES	YES
Observations	369,302	348,921	369,295	348,917	369,295	348,917
Number of diseases	9,863	9,504	9,863	9,504	9,863	9,504
Mean of the DV	0.0561	0.0564	0.0516	0.0518	0.0561	0.0564

Note: *, **,*** denote significance at the 5%, 1%, and 0.1% level, respectively. Cross-sectional regressions at the gene-disease association (GDA) level. Std. err. clustered at the gene and disease level. Columns (1)–(2) use the DisGeNET Score used in the main text. Columns (3)–(4) use an alternative Score that removes the contribution of post-discovery publications. Columns (5)–(6) return to the original Score but control for the amount of supporting evidence used to build it. *DisGeNET Score*: a synthetic measure of scientific reliability of the gene-disease association provided by DisGeNET. *DisGeNET Score excluding post-discovery literature*: the corrected Score excluding the contribution of publications appearing after the discovery year. *Evidence-volume controls*: categorical indicators for the number of publications (capped at 10) and the number of distinct non-literature sources.

D PubTator3 Data

This appendix provides additional details on PubTator3 and includes examples illustrating how its AI engine captures the epistemic nature of the gene–disease relationships investigated in scientific articles.

D.1 Details on PubTator3

PubTator3 is an advanced text-mining tool developed by the NIH's National Center for Biotechnology Information (NCBI) to help researchers extract biomedical concepts from scientific literature (Wei et al., 2024). It integrates natural language processing and state-of-the-art AI techniques to automatically recognize and normalize bio-entities (such as genes and diseases) and the relationships between them (e.g., a mutation in a gene causing a disease vs. merely co-occurring with it). PubTator3 currently provides entity and relation annotations across roughly 36 million PubMed abstracts and 6 million full-text articles from the PubMed Central open-access subset. The tool is freely accessible to the research community at: https://www.ncbi.nlm.nih.gov/research/pubtator3/.

PubTator3 employs a high-performance entity search engine that normalizes different forms of the same biological entity into standardized names, ensuring consistent classifications regardless of terminology variation. The system uses AIONER, an AI annotation tool, to identify six types of bio-entities: genes, diseases, chemicals, variants, species, and cell lines. Each entity type is normalized using specialized natural language processing modules tailored to existing terminology standards, such as GNormPlus for genes (mapped to NCBI Gene identifiers) and TaggerOne for diseases (mapped to MeSH terms). This normalization process ensures accurate and consistent identification across the literature. In addition, PubTator3 integrates BioREx, a transformer-based relation extraction model that identifies multiple types of relationships among bio-entities, including gene–disease interactions. BioREx significantly enhances relation extraction performance, achieving an F-score of 79.6% on standard test sets (Wei et al., 2024).

D.2 Entity Relationships from PubTator3

PubTator3 extracts 12 possible relationship types among bio-entities using BioREx. For the purposes of this paper, I classify the following PubTator3 relationship types as reflecting interactions of a *causal* nature:

- Gene A *causes* or *stimulates* disease B: when the status of one entity increases (or decreases) as the other increases (or decreases);
- Gene A *inhibits* or *prevents* disease B: when the status of one entity increases (or decreases) as the other decreases (or increases);

- Gene A *treats* or *co-treats* disease B: if a chemical or drug treats a disease, alone or in combination, through a given gene.

Note that all results remain robust when restricting the analysis to the strict relationship in which gene A *causes* disease B. Similarly, I classify the following PubTator3 relationship types as reflecting interactions of a *correlational* nature:

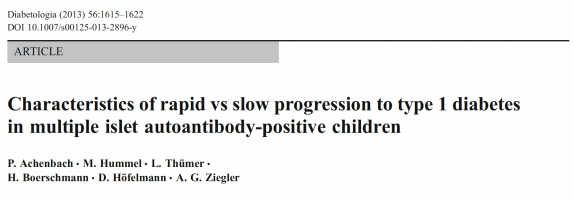
- Gene A *negatively correlates* with disease B: when the status of one entity tends to decrease (or increase) as the other increases (or decreases);
- Gene A *positively correlates* with disease B: when the status of one entity tends to increase (or decrease) as the other increases (or decreases);

I use the unique PubMed ID of each paper in DisGeNET to merge it with the corresponding relationships extracted by PubTator3. I retain only those relationships that involve either a causal or correlational link between the relevant genes and diseases. As a result, any paper in DisGeNET can be classified as presenting either causal or correlational relationships, based on how PubTator3 categorizes the gene–disease interactions it contains. Out of the 201,548 unique articles in my dataset, 74,634 include at least one causal relationship, 46,233 include at least one correlational relationship, and 29,432 include both (but involving different gene–disease pairs). Importantly, PubTator3 classifies the relationships described in scientific text, rather than independently verifying their causal validity. In this paper, I use follow-on articles classified as causal as a proxy for research aimed at understanding the mechanisms underlying gene-disease associations. This measure captures the extent to which subsequent scientific work engages with causal explanations, rather than whether those explanations have been conclusively validated or new theoretical knowledge has been established.

As an example, consider the GWAS by Hakonarson et al. (2007) that first reported the association between KIAA0350–type 1 diabetes. PubTator3 enables more detailed characterization of follow-on studies investigating the same gene-disease relationship. Panel (a) presents the title of a study by Achenbach et al. (2013), an observational study examining factors that influence the rate at which children with multiple islet autoantibodies develop type 1 diabetes. Among 1,650 children followed, 23 progressed to diabetes within 3 years (rapid progressors), while 24 remained non-diabetic for over 10 years (slow progressors). The study found that the presence or absence of several mutations, including in the gene KIAA0350, was correlated with the speed of disease progression. Accordingly, this paper is tagged in PubTator3 with the relationship

Figure D.1: Example of publications investigating correlational and causal relationships between the gene KIAA0350 and type 1 diabetes.

(a) *Paper on correlational relationship*


Diabetologia (2013) 56:1615–1622
DOI 10.1007/s00125-013-2896-y

ARTICLE

Characteristics of rapid vs slow progression to type 1 diabetes in multiple islet autoantibody-positive children

P. Achenbach · M. Hummel · L. Thümer · H. Boerschmann · D. Höfelmann · A. G. Ziegler

(b) *Paper on causal relationship*


Cell

The Diabetes Susceptibility Gene *Clec16a* Regulates Mitophagy

Scott A. Soleimanpour,[1,2] Aditi Gupta,[1] Marina Bakay,[3] Alana M. Ferrari,[1] David N. Groff,[1] João Fadista,[4] Lynn A. Spruce,[5] Jake A. Kushner,[6] Leif Groop,[4] Steven H. Seeholzer,[5] Brett A. Kaufman,[7] Hakon Hakonarson,[3,8,9] and Doris A. Stoffers[1,9,*]

Note: Panel (a) reports the title from Achenbach et al. (2013). Panel (b) reports the title from Soleimanpour et al. (2014). Clec16a is an alternative name for the KIAA0350 gene.

"gene KIAA0350 associates with disease diabetes mellitus".[4]

Panel (b) shows the title of the study by Soleimanpour et al. (2014), which investigates the mechanisms through which KIAA0350 influences the diseases it has been associated with, including type 1 diabetes. The researchers discovered that KIAA0350 encodes a protein that interacts with the enzyme Nrdp1, thereby protecting it from degradation. In turn, Nrdp1 regulates another key protein involved in the clearance of damaged mitochondria. These findings reveal KIAA0350's role in preserving mitochondrial health by regulating mitophagy, shedding light on the biological mechanisms by which this gene could emerge as a therapeutic target for diabetes. In PubTator3, this paper is tagged with the relationship "gene KIAA0350 inhibits disease diabetes mellitus." Notably, one of the authors of Soleimanpour et al. (2014) also co-authored the earlier GWAS by Hakonarson et al. (2007).

Finally, I use a case–control approach to further validate PubTator3's ability to capture the nature of gene–disease relationships studied in scientific articles. Specifically, one would expect clinical trials to be more likely to investigate causal relationships, given their objective of testing the causal effects of therapies. To explore this idea, I draw on data from NIH's iCite, which classifies an article as clinical if it is tagged with MeSH terms such as "Clinical Trial" or "Randomized Controlled Trial." Table D.1 confirms that articles reporting clinical trial results are significantly more likely to be classified as studying causal relationships in the PubTator3 data, and significantly less likely to present correlational evidence. This provides additional support for the validity of the PubTator3 classification.

[4]Interestingly, Achenbach et al. (2013) does not cite Hakonarson et al. (2007), even if it investigates the KIAA0350-type 1 diabetes association first introduced by it. This is one example of how my measurement approach can better capture the impact of combinations by not relying on paper-to-paper citations.

Table D.1: Clinical articles are more likely to explore causal relationships between a gene and a disease, and less likely to explore correlational relationships.

	Causal relationship between gene and disease (0/1)		Correlational relationship between gene and disease (0/1)	
	(1)	(2)	(3)	(4)
Clinical article (0/1)	0.0684***	0.0477**	-0.0443***	-0.0145
	(0.0129)	(0.0150)	(0.00999)	(0.0110)
Disease FE	YES	YES	YES	YES
Year of discovery FE	YES	YES	YES	YES
Principal investigator FE	NO	YES	NO	YES
Observations	369,305	347,776	369,305	347,776
Number of diseases	9,864	9,494	9,864	9,494
Mean of the DV	0.386	0.386	0.246	0.246

Note: *, **,*** denote significance at the 5%, 1%, and 0.1% level, respectively. Cross-sectional regressions at the gene-disease association (GDA) level. Std. err. clustered at the gene and disease level. *Causal relationship between gene and disease (0/1)*: 0/1 = 1 if the article introduces at least one causal relationship about a gene-disease pair according to PubTator3 (Wei et al., 2024); *Correlational relationship between gene and disease (0/1)*: 0/1 = 1 if the article introduces at least one correlational relationship about a gene-disease pair according to PubTator3 (Wei et al., 2024); *Clinical article (0/1)*: 0/1 = 1 for new gene-disease associations introduced by an article reporting results of a clinical trial (data on clinical articles from NIH's iCite).

E Anomaly Index

This appendix provides details on the construction of the Anomaly Index for each novel gene–disease association. It begins with a stylized example illustrating the measure's logic, which is computed based on established patterns of genetic co-occurrence, followed by a more detailed explanation and a set of robustness checks.

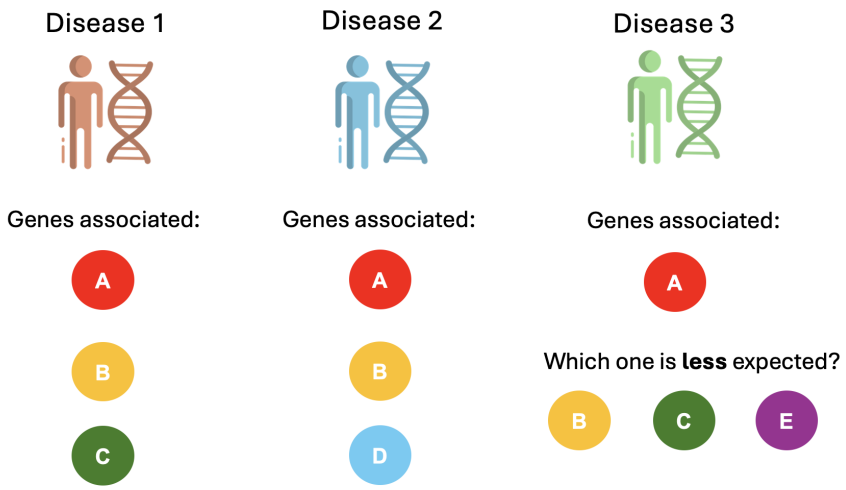
E.1 Construction and Stylized Example

Consider the following example. There are three diseases:

- Disease 1, which has been associated with genes $\{A, B, C\}$ at time t;
- Disease 2, which has been associated with genes $\{A, B, D\}$ at time t;
- Disease 3, which has been associated with gene $\{A\}$ at time t.

Suppose now that Disease 3 can potentially be associated with genes B, C, and E in time $t + 1$. How anomalous would each potential combination between these genes and Disease 3 be, based on pre-existing association patterns?

Figure E.1: Example of how the Anomaly Index for each new gene-disease association is computed.



Note: See text for the step-by-step computation based on this stylized example.

The first step involves computing the frequency of gene co-occurrences from past discoveries. In this example, we have the pairs of genes $\{(A, B), (A, C), (B, C)\}$ from Disease 1 and $\{(A, B), (A, D), (B, D)\}$ from Disease 2. Counting in how many diseases each pair occurs gives us the following frequencies:

- $\text{freq}(A, B) = 2$ (appears in Diseases 1 and 2)

- $\text{freq}(A, C) = 1$ (appears only in Disease 1)
- $\text{freq}(B, C) = 1$ (appears only in Disease 1)
- $\text{freq}(A, D) = 1$ (appears only in Disease 2)
- $\text{freq}(B, D) = 1$ (appears only in Disease 2)
- Any other potential pair (e.g. $(C, D), (A, E)$) has frequency 0.

Against this backdrop, Disease 3 has only been associated with the gene $\{A\}$ up to time t. Based on the pattern of genetic co-occurrences in Diseases 1 and 2, the association between Disease 3 and gene $\{A\}$ allows me to quantify how strongly prior genetic knowledge would anticipate each potential new gene link for Disease 3 at time $t + 1$. Accordingly, the *expectation* of linking a generic gene g to Disease 3, given its existing gene set $\{A\}$, is:

$\text{Expectation}(g, \text{Disease 3}) = \sum_{x \in \{A\}} \text{freq}(g, x) = \text{freq}(g, A)$

Which then leads to the following:

- $\text{freq}(B, A) = 2 \implies \text{Expectation}(B, \text{Disease 3}) = 2$
- $\text{freq}(C, A) = 1 \implies \text{Expectation}(C, \text{Disease 3}) = 1$
- $\text{freq}(E, A) = 0 \implies \text{Expectation}(E, \text{Disease 3}) = 0$

From this, I define the Anomaly Index as:

$\text{Anomaly Index}(g) = \frac{1}{1+\text{Expectation}(g)}$,

thus leading to the following calculations:

$$\text{Anomaly Index}(B, \text{Disease 3}) = \frac{1}{1+2} = \frac{1}{3} \approx 0.33,$$
$$\text{Anomaly Index}(C, \text{Disease 3}) = \frac{1}{1+1} = \frac{1}{2} = 0.50,$$
$$\text{Anomaly Index}(E, \text{Disease 3}) = \frac{1}{1+0} = 1.0.$$

Intuitively, the Anomaly Index ranges from 0 to 1, with higher values indicating that a given gene–disease pair is more discordant with prior discovery patterns observed in other diseases. The index is undefined for the first gene ever linked to a disease. It also equals 1 by construction when the gene has never been previously associated with any disease, as with gene E in the example above.

E.2 Validation of the Anomaly Index

Generalizing the construction above, let $G_{d,t-1}$ denote the set of genes associated with disease d before year t, and let $f_{gh,t-1}$ denote the number of diseases with which genes g and h had both been associated before

that year. For a newly reported association between gene g and disease d, I define:

$$\text{Anomaly Index}_{gd,t} = \frac{1}{1 + \sum_{h \in G_{d,t-1}} f_{gh,t-1}}. \tag{E.1}$$

The Index takes values in $(0, 1]$ and is higher when the focal gene has fewer recorded co-occurrences with genes already associated with the disease. Applying this procedure to gene-disease combinations introduced during the study period, I calculated the Index for 98.4% of the gene–disease pairs in my sample; the remaining 6,040 pairs involve the first gene ever associated with a given disease, for which the Index is not defined. The Anomaly Index reaches a maximum of 1 for the 11,289 combinations that include a gene that has never been previously linked to any disease. Consider the studies by Hakonarson et al. (2007) and Qu et al. (2007), both published in 2007 and focused on type 1 diabetes—a disease that, by the end of 2006, had already been associated with 1,020 genes. The Anomaly Index for the KIAA0350–type 1 diabetes combination is 1, since it marked the first time KIAA0350 had been linked to any disease. In contrast, the Anomaly Index for the IRF5–type 1 diabetes combination is 0.00077, indicating that IRF5 had frequently co-occurred with genes already associated with diabetes up to that point.[5]

Several additional analyses support the face validity of the Anomaly Index. First, I link the Index to how authors describe their findings in the abstract of the originating publication. Using data by Mishra et al. (2023), who apply a probabilistic model to classify adjectives in abstracts, I show that papers that label their discovery as "novel" introduce gene–disease pairs with an Anomaly Index that is 40–46% higher on average (Table E.1). By contrast, I find no comparable relationship for terms such as "major" or "critical," which suggests that the result is not driven by generic hype in the abstract.

Second, I benchmark the Anomaly Index against expert assessments from the Faculty Opinions platform, following Shi and Evans (2023). Faculty Opinions provides post-publication evaluations in which domain experts annotate papers using standardized tags, including whether a study reports a "new finding" or a "technical advance." These annotations are available for 6,275 of the papers in my sample that introduce 11,465 new gene–disease associations. Table E.2 shows that, in this subset, the Anomaly Index is positively and significantly associated with papers tagged as a "new finding," but not with tags capturing other forms of novelty, such as the use of a new technique. Interestingly, I find no statistically significant relationship with the "controversial" tag. Thus, while these results support interpreting the Anomaly Index as capturing novel and unexpected findings, they do not establish that such discoveries challenge existing scientific paradigms.

[5] Sparse prior research can also yield high Anomaly Index values. Panel B of Table G.10 examines whether GWAS discoveries remain more anomalous after accounting for prior publication volume and prior disease associations, along with gene and disease-by-year fixed effects. The estimated GWAS coefficient becomes smaller but remains positive and statistically significant.

Table E.1: The Anomaly Index is higher for gene-disease combinations described as "novel" in the paper abstract, but not for other words generically denoting hype

	Anomaly Index of Novel Gene-Disease Associations					
	(1)	(2)	(3)	(4)	(5)	(6)
I(Results described as "Novel">0)	0.0123***	0.0105***				
	(0.00144)	(0.00159)				
I(Results described as "Major">0)			-0.000851	-0.000852		
			(0.00124)	(0.00178)		
I(Results described as "Critical">0)					0.000171	-0.000130
					(0.00125)	(0.00170)
Disease FE	YES	YES	YES	YES	YES	YES
Year of discovery FE	YES	YES	YES	YES	YES	YES
Principal investigator FE	NO	YES	NO	YES	NO	YES
Observations	365,340	345,040	365,340	345,040	365,340	345,040
Number of diseases	8,746	8,450	8,746	8,450	8,746	8,450
Mean of the DV	0.027	0.027	0.027	0.027	0.027	0.027

Note: *, **,*** denote significance at the 5%, 1%, and 0.1% level, respectively. Cross-sectional regressions at the gene-disease association (GDA) level. Std. err. clustered at the disease level. *Anomaly Index*: a synthetic measure of how unlikely a given gene was to be associated with a disease given the pattern of genes previously associated with that disease up to the year before; *I(Results described as "Novel">0)*: 0/1 = 1 if the gene-disease association is described as "novel" in the abstract of the paper reporting it, according to the data by Mishra et al. (2023); *I(Results described as "Major">0)*: 0/1 = 1 if the gene-disease association is described as "major" in the abstract of the paper reporting it, according to the data by Mishra et al. (2023); *I(Results described as "Critical">0)*: 0/1 = 1 if the gene-disease association is described as "critical" in the abstract of the paper reporting it, according to the data by Mishra et al. (2023).

Table E.2: The Anomaly Index is higher for gene-disease combinations rated as "new findings", but not for other types of ratings by scientists on Faculty Opinions.

	Anomaly Index			
	(1)	(2)	(3)	(4)
I(Rated as "New Finding">0)	0.0877**			
	(0.0267)			
I(Rated as "New Drug Target">0)		0.0209		
		(0.0148)		
I(Rated as "Controversial">0)			-0.0163	
			(0.0266)	
I(Rated as "New Technique">0)				-0.0119
				(0.0215)
Disease FE	YES	YES	YES	YES
Year of discovery FE	YES	YES	YES	YES
Principal investigator FE	YES	YES	YES	YES
Observations	11,465	11,465	11,465	11,465
Number of diseases	1,471	1,471	1,471	1,471
Mean of the DV	0.057	0.057	0.057	0.057

Note: *, **,*** denote significance at the 5%, 1%, and 0.1% level, respectively. Cross-sectional regressions at the gene-disease association (GDA) level. Std. err. clustered at the gene and disease level. *Anomaly Index*: a synthetic measure of how unlikely a given gene was to be associated with a disease given the pattern of genes previously associated with that disease up to the year before; *I(Results rated as "New Finding">0)*: 0/1 = 1 if the gene-disease association is rated as a "new finding" by scientists in the Faculty Opinions platform; *I(Results rated as "New Drug Target">0)*: 0/1 = 1 if the gene-disease association is rated as constituting a "new drug target" by scientists in the Faculty Opinions platform; *I(Results rated as "Controversial">0)*: 0/1 = 1 if the gene-disease association is rated as being "controversial" by scientists in the Faculty Opinions platform; *I(Results rated as "New Technique">0)*: 0/1 = 1 if the gene-disease association is rated as employing a "new technique" by scientists in the Faculty Opinions platform.

F Details on the Instrumental Variable Strategy

F.1 Endogeneity Concerns

The main findings indicate that GWAS discoveries present systematically different characteristics than those uncovered by candidate gene studies. However, since GWAS adoption is a researcher's choice, it may be correlated with unobserved determinants of discovery outcomes. Principal investigator (PI) fixed effects strengthen identification by comparing outcomes for the same scientist before and after switching search strategies, but bias could persist if researchers with higher or lower latent ability are more likely to adopt GWAS. The sign of this bias is unclear ex ante. If scientists adopt GWAS after struggling to discover valuable gene-disease associations through theory-driven search, estimates would be biased downward. If, instead, the most capable scientists are early adopters, estimates would be biased upward. Appendix Figure G.1 helps address this concern by showing that, prior to their first GWAS paper, eventual adopters do not systematically discover gene-disease associations of different quality.

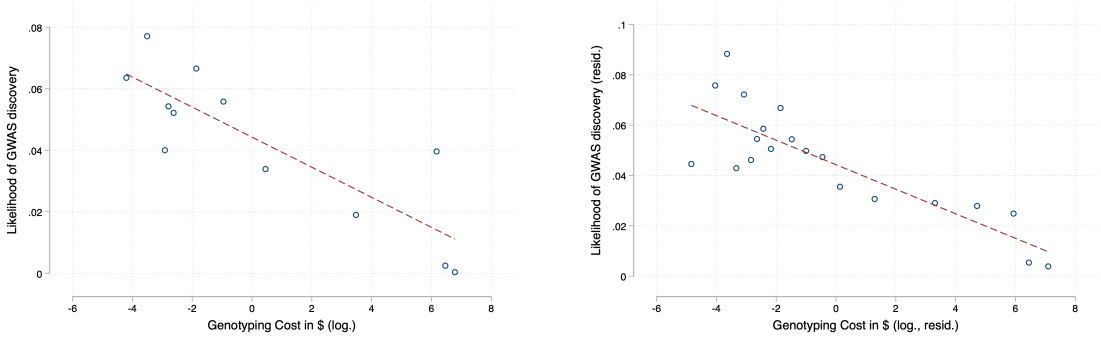
A more serious concern regarding identification is that the timing of GWAS adoption may be influenced by unobserved, time-varying factors that also impact outcomes. At the PI level, shifts in research resources or scientific standing could increase the likelihood of adopting GWAS while simultaneously shaping the quality and impact of their discoveries. At the disease level, adoption incentives may depend on the state of prior knowledge, since diseases with well-developed theory may offer less apparent value from data-driven search. Even with disease and disease-by-year fixed effects, OLS estimates may still be exposed to disease-specific shocks that jointly influence the adoption of GWAS and its outcomes. To address these concerns, the remainder of this appendix introduces and validates an instrumental variable strategy (IV) that isolates plausibly exogenous variation in GWAS adoption and provides a complementary test of the baseline estimates.

F.2 Exploiting the Declining Costs of Large-Scale Genetic Data

GWAS are carried out using genotyping microarrays (or chips), that is, physical slides with millions of synthetic DNA probes that match specific genetic variations (SNPs) in the human genome. When GWAS first emerged around 2005, the cost of conducting a single study was estimated at $10 million or more (Visscher et al., 2012). Over time, genome-wide genotyping became markedly more affordable thanks to the diffusion of cheaper microarrays (Panel (a) of Figure 3). I exploit this technology-driven decline as a supply-side shock to the feasibility of GWAS. Specifically, I use NIH data on the average cost of collecting one megabase (i.e., one million bases) of DNA in a given year. Data are available at: `www.genome.gov/sequencingcostsdata`.

Although this series measures sequencing rather than microarray genotyping costs directly, it captures the broader decline in the cost of producing large-scale genetic data. Consistent with this logic, Figure F.1 shows that the share of new gene–disease associations discovered in year t that are identified via GWAS declines as genotyping costs in year t increase. Lower costs thus increase the likelihood of adopting GWAS, providing the first ingredient of the IV strategy.

Figure F.1: Lower genetic-data costs are associated with greater use of GWAS in new discoveries.



(a) Raw correlation between genotyping costs and GWAS use

(b) Conditional correlation between genotyping costs and GWAS use

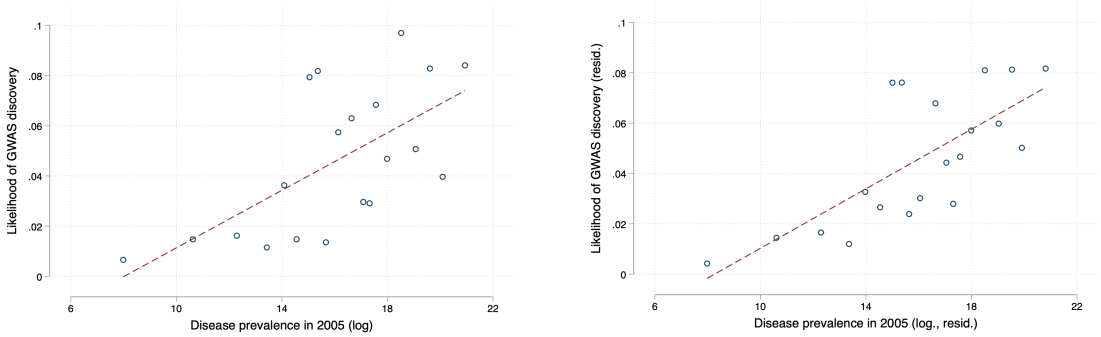
Note: The figure plots binned correlations between the genotyping cost in a given year and the likelihood that a gene-disease association first reported in that year is introduced via GWAS. Panel (a) reports the unconditional relationship. Panel (b) reports the relationship after residualizing the outcome by disease.

While the reduction in genotyping costs was a common trend, the diffusion of GWAS was uneven across diseases. In particular, more common diseases experienced a larger increase in GWAS use (Panel b of Figure 3). Two forces help explain this heterogeneity. On the demand side, funding agencies explicitly prioritized "common complex diseases" with high public health salience (Heeney, 2021). For example, the Wellcome Trust targeted the first GWAS to "seven diseases of major public health importance," such as diabetes and coronary heart disease (WTCCC, 2007). On the supply side, GWAS required assembling large case-control samples, making recruitment more feasible when patient pools were abundant. The Wellcome Trust noted that its early focus was shaped by the fact that "suitable nationally representative sample sets were available" (WTCCC, 2007), which was obviously easier for diseases with higher prevalence in the population. Together, these forces meant that declining genotyping costs lowered the barriers to GWAS adoption more substantially for common diseases, providing the basis for my IV strategy.

To capture how the same cost decline translated into different adoption responses across diseases, I construct a measure of disease prevalence in the pre-GWAS period. Prevalence is the number of individuals living with a given disease at a point in time, reflecting the overall burden of the condition. In the context of my

identification strategy, it proxies both for the prioritization of more important diseases (a pull factor) and for the ease of assembling the large samples required for GWAS (a push factor). I measure prevalence as of 2005, immediately before GWAS diffusion, so it is predetermined with respect to subsequent GWAS adoption and discovery outcomes. I obtain prevalence information from two sources. First, for diseases covered by the Global Burden of Disease (GBD) study, I use the 2023 estimates of the number of prevalent cases in 2005, covering all ages, both sexes, and the global population. To link these estimates to the diseases in my sample, I map UMLS disease identifiers to ICD-10-CM codes using the UMLS Metathesaurus (MRCONSO.RRF), and then to GBD disease categories using the official IHME mappings. Second, I use Orphanet, which reports disease frequency for rare conditions. In total, prevalence information is available for 2,511 diseases, covering 89,368 gene-disease associations discovered after 2005, of which 4,064 are introduced via GWAS.[6] Using these data, I assign each disease a pre-GWAS prevalence measure and hold it fixed at its 2005 value throughout the IV analysis. Figure F.2 shows that GWAS use is higher in disease areas with greater prevalence, even after controlling for year. This provides the second ingredient of the IV strategy by capturing cross-disease differences in the feasibility and incentives to adopt GWAS.

Figure F.2: Higher disease prevalence is associated with greater use of GWAS in new discoveries.



(a) Raw correlation between 2005 disease prevalence and GWAS use

(b) Conditional correlation between 2005 disease prevalence and GWAS use

Note: The figure plots binned correlations between disease prevalence measured in 2005 and the likelihood that a gene-disease association first reported in year t is introduced via GWAS. Panel (a) reports the unconditional relationship. Panel (b) reports the relationship after residualizing the outcome by year.

F.3 Construction of the IV

To build the instrument, I combine the two elements discussed above to capture the idea that the exogenous decrease in genotyping costs had heterogeneous effects across diseases with different baseline prevalence.

[6]The estimation sample in Table 4 is further reduced because diseases with a single observed discovery are dropped when including fixed effects. Appendix Table G.7 shows that the main OLS results remain robust when replicated on the smaller sample used for the IV analysis.

Following Borusyak et al. (2025), this yields a simplified shift share structure:

$$\text{Genotyping cost shift-share}_{j,t} = \underbrace{\text{Genotyping Cost}_t}_{\text{Shift}} \times \underbrace{\text{Disease Prevalence}_{j,2005}}_{\text{Share}}, \tag{F.1}$$

where the "shift", $Genotyping\ Cost_t$, is a common time series capturing the decline in the cost of collecting genetic data, and the "share", $Disease\ Prevalence_{j,2005}$, is a predetermined measure of disease j's prevalence in the population. Because there is a common time-varying shock, the baseline prevalence can be interpreted as a disease-level exposure, capturing how strongly each disease is predicted to respond to the same exogenous change in genotyping costs. The identifying variation, therefore, comes from differential adoption responses across diseases with different baseline prevalence as genotyping costs fall.

The first stage is estimated at the discovery level and relates the probability that discovery i for disease j in year t is introduced via GWAS to the shift-share instrument, controlling for the vector of baseline covariates in $\mathbf{Z}_{i,j,t}$ and fixed effects described in the main text:

$$\text{GWAS (0/1)}_{i,j,t} = \pi_0 + \pi_1 \text{ Genotyping cost shift-share}_{j,t} + \mathbf{Z}'_{i,j,t}\pi + u_{i,j,t}. \tag{F.2}$$

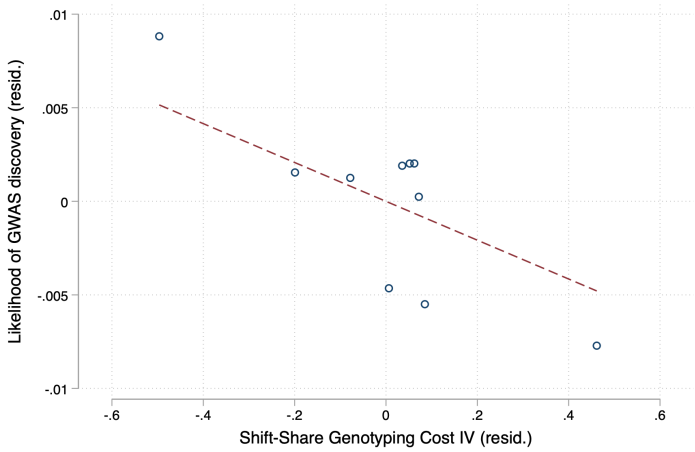
The instrument varies at the disease-year level, since it depends on the prevalence of disease j measured in 2005 and genotyping costs in year t, thereby shifting GWAS adoption through differential exposure to an exogenous cost shock. Standard errors are clustered two ways, at the gene and disease-class levels. Clustering at the broader disease-class level allows for correlated unobserved shocks across related diseases. This is particularly relevant in the smaller IV sample, where several diseases may share common research conditions or changes in funding.

Empirical estimates show that the first stage is strong: higher values of the instrument, which correspond to years with higher genotyping costs, are associated with a lower likelihood that a discovery is introduced via GWAS ($\hat{\pi}_1 = -0.0104$, s.e. $= 0.00235$, $p = 0.004$). Figure F.3 visualizes the first stage relationship. The instrument is highly relevant, with Cragg Donald Wald $F = 53.6$ and Kleibergen-Paap rk Wald $F = 19.5$.

F.4 Validation of the IV

In this section, I follow recent econometric work on shift-share instruments to probe the validity of the proposed IV (Goldsmith-Pinkham et al., 2020; Borusyak et al., 2025). Here, I focus on the suitability of the shares, namely disease prevalence measured in 2005 (before GWAS emergence), as a disease-level exposure measure to the common exogenous shock generated by changes in genotyping costs. Identification relies on

Figure F.3: First stage: higher values of the shift-share genotyping costs IV predict lower GWAS adoption.

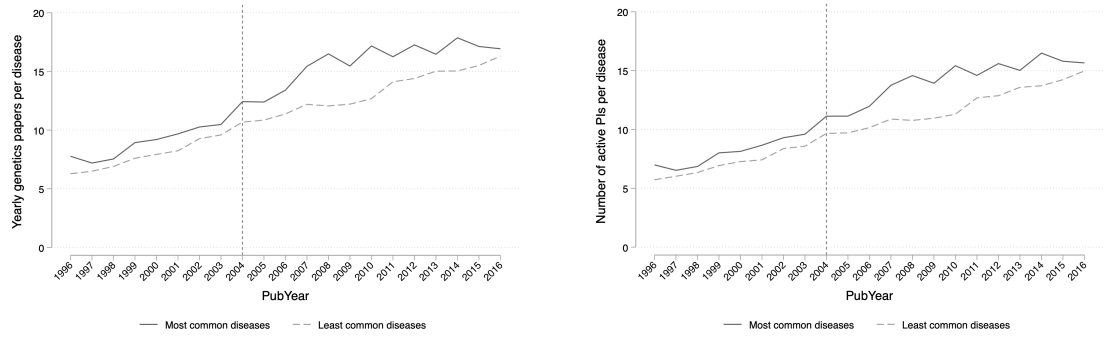


Note: The figure plots a binned relationship between the genotyping cost shift-share instrument and the likelihood that a gene disease association first reported in year t is introduced via GWAS. The estimates are residualized by disease, year, and PI fixed effects.

a "parallel trends type" assumption with respect to the shares: absent the cost decline, high-prevalence and low-prevalence diseases would have exhibited similar trajectories in GWAS adoption and in the discovery outcomes studied, conditional on the fixed effects and controls included in the empirical specification. Below, I follow the checklist proposed by Borusyak et al. (2025) to assess these conditions in my context.

1. Conceptual suitability: Disease prevalence measured in 2005 is a suitable exposure share because it is predetermined relative to GWAS diffusion and it captures a disease's differential sensitivity to the common decline in genotyping costs. When genome-wide data collection becomes cheaper, the adoption response should be stronger precisely where GWAS is most feasible and most valuable: diseases with larger potential case pools for assembling high-powered samples and greater public health salience that attracts early funding (Figure F.2). Figure F.4 provides additional descriptive evidence by plotting long-run trends in scientific activity for diseases with higher versus lower baseline prevalence. Reassuringly, these series show no evidence of differential pre-trends in research activity, with divergence emerging only after GWAS diffusion. However, this evidence cannot fully rule out the possibility that high-prevalence diseases attracted greater research funding or other resources over the same period, independently of GWAS adoption. Although I find no evidence in the available sources of such differential changes, this remains a potential alternative channel that would violate the exclusion restriction.

Figure F.4: Scientific activity evolves similarly across high and low prevalence diseases prior to GWAS diffusion.



(a) Publications on a disease

(b) PIs researching a disease

Note: The figure plots descriptive trends in scientific activity for diseases above versus below the median prevalence measured in 2005. Panel (a) reports the yearly number of publications in each group. Panel (b) reports the yearly number of active principal investigators in each group.

2. Unit-level controls: The shift-share instrument varies only through the interaction between baseline prevalence and annual genotyping costs. The empirical specifications include two sets of unit-level controls that isolate this interaction from obvious confounds. Year fixed effects absorb any shocks common to all diseases in a given year, including changes in the funding environment. Disease fixed effects absorb all time-invariant differences across diseases, such as baseline research intensity or intrinsic difficulty. With these fixed effects, the instrument does not compare high- versus low-prevalence diseases in levels. Instead, identification comes from comparing how outcomes within the same disease evolve as genotyping costs fall, with the predicted adoption response scaled by that disease's prevalence in 2005.

3. Characterizing which shares matter for identification: With a single exposure variable interacted with a common aggregate shock, the shift-share design does not admit the standard Rotemberg weights decomposition (Goldsmith-Pinkham et al., 2020). Instead, I characterize which diseases contribute most to the first-stage covariance and assess how sensitive the IV estimates are to their exclusion. Let n index gene–disease associations in the estimation sample, with $j(n)$ and $t(n)$ denoting the disease and discovery year of observation n. The instrument is $G_n = Prevalence_{j(n),2005} \times GenotypingCost_{t(n)}$, and the endogenous regressor is $D_n = \text{GWAS}_n$. Let $\widetilde{G}_n$ and $\widetilde{D}_n$ denote the residuals of these variables after partialling out the controls and fixed effects used in the main specification. Instrument relevance depends on the residualized first-stage cross-product $\sum_n \widetilde{G}_n \widetilde{D}_n$, which admits an exact decomposition by disease. Define each disease's contribution as $C_j = \sum_{n:j(n)=j} \widetilde{G}_n \widetilde{D}_n$, so that $\sum_j C_j = \sum_n \widetilde{G}_n \widetilde{D}_n$, and the corresponding relative contribution as $w_j = C_j / \sum_{j'} C_{j'}$. These weights sum to one and measure each disease's contribution to

the residualized first-stage cross-product, although individual weights can be negative.

Column (2) of Table F.1 reports these weights. The five largest contributors account for approximately 35% of the total, indicating that a subset of diseases contributes disproportionately to instrument relevance. To assess whether these diseases also drive the IV estimates, I implement a "leave-one-disease-out" sensitivity analysis. For each disease j, I re-estimate the 2SLS specifications after excluding all discoveries associated with that disease and obtain the corresponding coefficient $\hat{\beta}_{-j}$. Columns (3) through (6) of Table F.1 report these estimates for the ten diseases with the largest first-stage contributions. Reassuringly, the coefficients remain positive across these exclusions, although their magnitudes vary, particularly for scientific quality and follow-on research. These results suggest that the main findings are not driven entirely by any single disease among the largest contributors, while indicating some sensitivity in the magnitude of the IV estimates.

Table F.1: Robustness check: diseases that contribute most to IV identification and sensitivity to their exclusion.

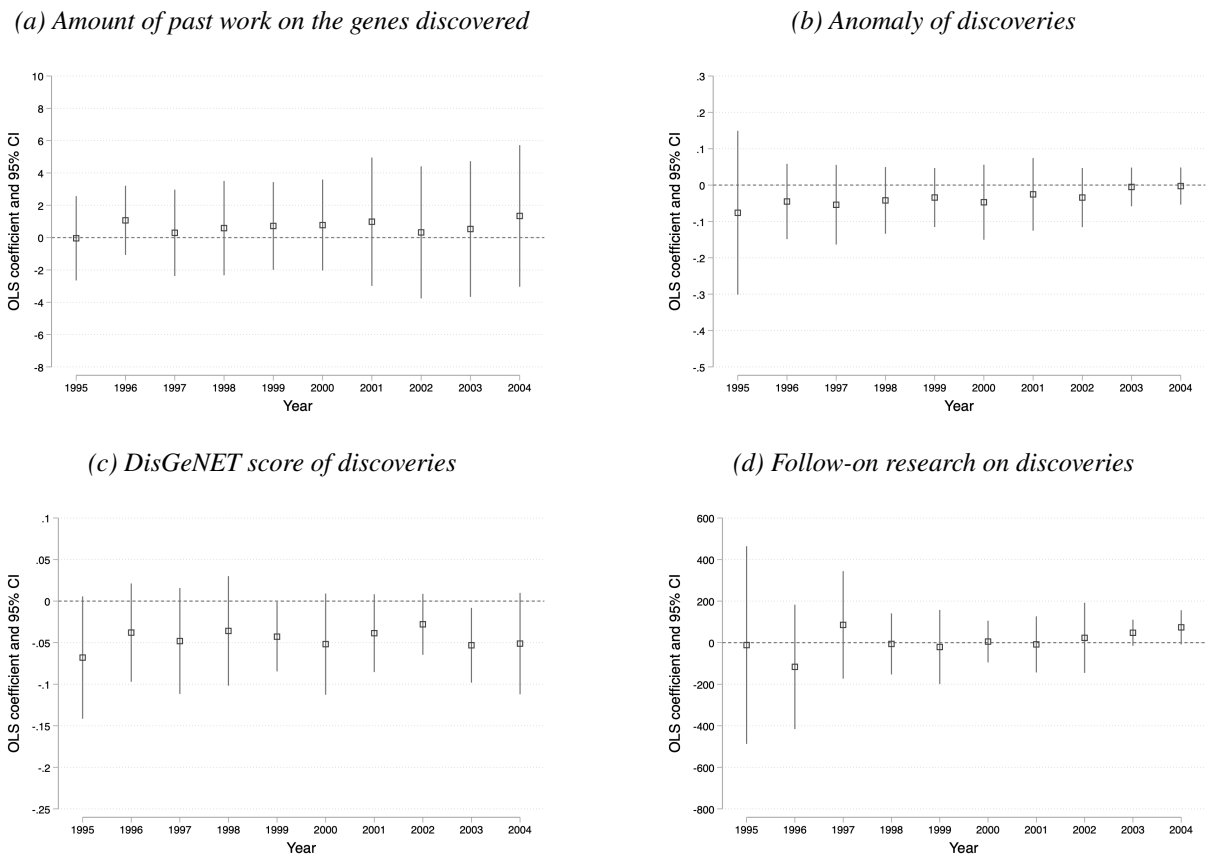
ID	Disease Name	GDAs (1)	Weight (2)	$\hat{\beta}_{-j}^{(IV,U)}$ (3)	$\hat{\beta}_{-j}^{(IV,D)}$ (4)	$\hat{\beta}_{-j}^{(IV,S)}$ (5)	$\hat{\beta}_{-j}^{(IV,F)}$ (6)
2774	Migraine Disorders	191	0.110	0.504	0.359	0.187	3.198
202	Asthma	928	0.088	0.468	0.258	0.217	4.752
586	Dental caries	75	0.080	0.522	0.224	0.163	1.979
231	Barrett Esophagus	263	0.043	0.527	0.279	0.118	1.892
5676	Common Migraine	48	0.034	0.502	0.220	0.142	3.087
604	Atopic Dermatitis	366	0.025	0.481	0.250	0.154	4.831
1425	Systemic Lupus Erythematosus	836	0.022	0.460	0.246	0.147	2.685
2149	Systemic Scleroderma	373	0.015	0.474	0.268	0.141	2.673
998	Sensorineural Hearing Loss	119	0.015	0.365	0.246	0.160	2.358
619	Type 1 Diabetes	634	0.015	0.467	0.263	0.163	3.249

Note: This table reports diagnostics for the genotyping costs shift-share IV. Column (1) reports the number of post-2005 gene-disease associations involving disease j. Column (2) reports the disease contribution weight w_j computed from the residualized first stage, as explained in the text above. Columns (3)–(6) report leave-one-disease-out 2SLS coefficients, $\hat{\beta}_{-j}^{IV}$, for the main outcomes: introduction of gene-disease associations (GDAs) including a gene never studied before (U) or recently discovered (D), DisGeNET score of the GDA (S), and follow-on publications on the GDA (F). The full-sample 2SLS coefficients are reported in the corresponding main IV tables.

4. Balance tests of the shares: Although the exclusion restriction cannot be tested directly, recent work on shift-share designs emphasizes balance tests as a plausibility check (Borusyak et al., 2025). The logic is that if baseline prevalence is a valid exposure share, then before GWAS emerges and genotyping costs decline, diseases with higher versus lower prevalence should not display systematically different trends in the outcomes studied. In this setting, this corresponds to the parallel-trends component of the identifying assumption: absent the cost decline, discovery outcomes in high-prevalence and low-prevalence diseases

would have evolved similarly, conditional on the fixed effects and controls in the empirical specification. Figure F.5 implements this diagnostic by comparing pre-2005 trends in the main dependent variables for diseases in the top and bottom prevalence quartiles in 2005. Across outcomes, there is no evidence of differential pre-trends prior to GWAS diffusion. This pattern supports interpreting baseline prevalence as an appropriate exposure share for the common cost shock.

Figure F.5: No evidence of pre-trends in discovery outcomes for high- versus low-prevalence diseases.



Note: The figure plots average outcomes by calendar year separately for diseases in the top versus bottom quartile of prevalence measured in 2005. The pre-2005 period provides a balance-style diagnostic for the share-based identifying assumption: absent the decline in genotyping costs and the diffusion of GWAS, outcomes for high- and low-prevalence diseases should display similar trends, conditional on controls.

5. Additional concerns about the exclusion restriction: A potential concern is that declining genotyping costs may have improved candidate gene studies independently of GWAS adoption. If cheaper genotyping enabled larger samples and better discoveries, particularly in high-prevalence diseases, the instrument could capture improvements in the research quality of candidate gene studies. Differential improvements in candidate gene studies would represent a potential violation of the exclusion restriction. However, while a theoretical possibility, this risk seems unlikely because candidate gene studies genotype a limited number

of variants by design, and therefore incur substantially lower genotyping costs than genome-wide studies. Moreover, reducing genotyping costs does not necessarily relax other constraints on sample size, such as patient recruitment and phenotyping.

Nevertheless, I examine this possibility using data on study sample sizes. Unlike GWAS, candidate gene studies are not covered by a comprehensive database reporting sample sizes. I therefore assemble the available information from the Genetic Association Database (GAD) (Becker et al., 2004) and the historical candidate gene studies documented by Farrell et al. (2015). These sources provide sample-size information for a limited subset of diseases. Table F.2 shows no statistically significant increase in candidate gene study sample sizes over time, with or without disease fixed effects. By contrast, sample sizes increased substantially for GWAS studies, consistent with the logic of the IV.

Table F.2: The average sample size of GWAS grows over time, as genotyping costs decrease, but the same is not true for candidate gene studies.

	Total Number of Individuals			
	Candidate-Gene Sources		GWAS	
	(1)	(2)	(3)	(4)
Publication Year	-53.256	-54.192	3,849.828**	4,438.265***
	(121.972)	(162.346)	(1,140.587)	(644.289)
Disease FE	NO	YES	NO	YES
Observations	8,559	8,559	2,415	1,977

Note: *, **, *** denote significance at the 5%, 1%, and 0.1% levels, respectively. The dependent variable is the total number of individuals included in a study. Columns (1)–(2) use the data from Becker et al. (2004) and Farrell et al. (2015). Columns (3)–(4) use data from the GWAS Catalog. The independent variable is publication year. Standard errors are clustered at the disease and publication-year levels.

Finally, the 2SLS estimates in the main text are generally larger than their OLS counterparts. This difference likely reflects treatment-effect heterogeneity: the IV identifies a local average treatment effect (LATE) for discoveries whose research approach responds to declining genotyping costs, which need not coincide with the average effect. However, larger IV estimates could also signal violations of the exclusion restriction. In particular, high-prevalence diseases may have attracted increasing funding, researchers, or other resources independently of GWAS adoption, potentially improving discovery outcomes through alternative channels. Although the preceding diagnostics provide some reassurance, they cannot fully rule out these possibilities.

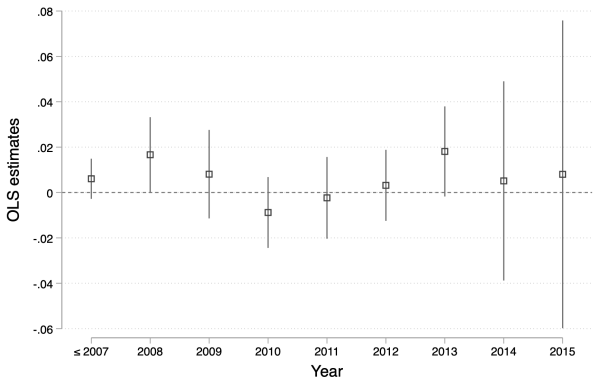
Overall, the evidence supports the validity of the IV strategy. The instrument exploits a technology-driven cost shock that predicts GWAS adoption, and its heterogeneity is anchored in a predetermined measure of baseline prevalence that captures differential exposure to cost reductions. Consistent with the identifying

logic, the balance and sensitivity exercises do not reveal patterns that would suggest systematic confounding. This strengthens the interpretation of the 2SLS estimates as reflecting variation in search strategy induced by the decline in genotyping costs.

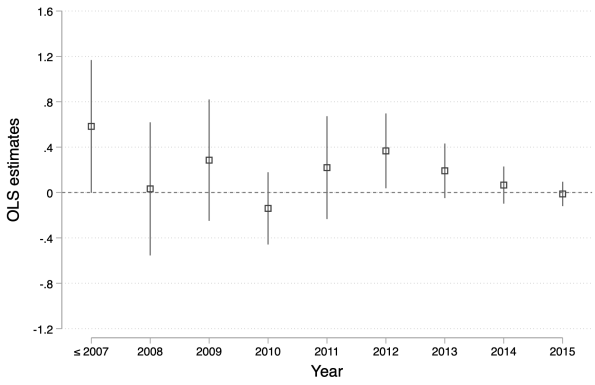
G Additional Figures and Tables

Figure G.1: Robustness check: Before their first GWAS, principal investigators who adopt data-driven search are not systematically more likely to introduce gene-disease associations of higher value or to receive more follow-on work.

(a) Pre-GWAS average DisGeNET Score of discoveries by future GWAS adopters



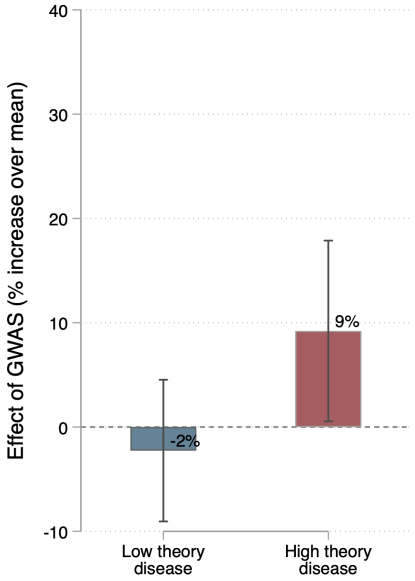
(b) Pre-GWAS average follow-on studies for discoveries by future GWAS adopters



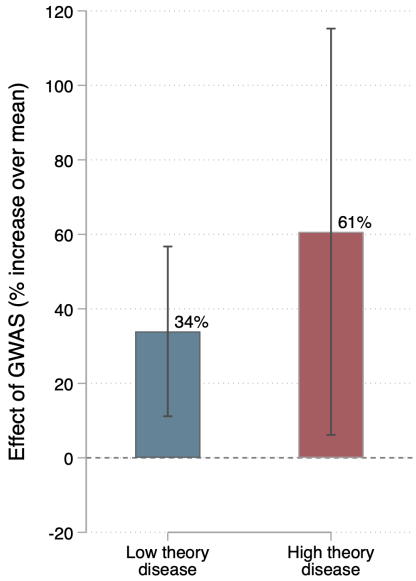
Note: This figure investigates if there are systematic differences in the scientific quality of gene-disease combinations introduced by scientists that self-select into publishing GWAS. Every coefficient is estimated from a separate regression for a given year, in which I compare PIs who never publish a GWAS with PIs who will eventually publish one but have not yet done so (that is, PIs are excluded from the sample after their first GWAS). In practice, each β coefficient is estimated from the following specification: $(Feature\ of\ gene\text{-}disease\ combinations\ in\ year\ t)_{i,j} = \alpha + \beta(PI\ will\ publish\ GWAS)_{i,j,t} + \mathbf{Z}'_{i,j}\gamma + \epsilon_{i,j,t}$. Panel (a) plots the coefficients and confidence intervals for the DisGeNET score of GDAs introduced in a given year. Panel (b) plots the coefficients and confidence intervals for the average count of follow-on papers received by GDAs introduced in a given year.

Figure G.2: Heterogeneity in GWAS discovery quality across diseases with more versus less pre-existing genetic knowledge.

(i) I(GDA in bottom 50% DisGeNET Score > 0*)*

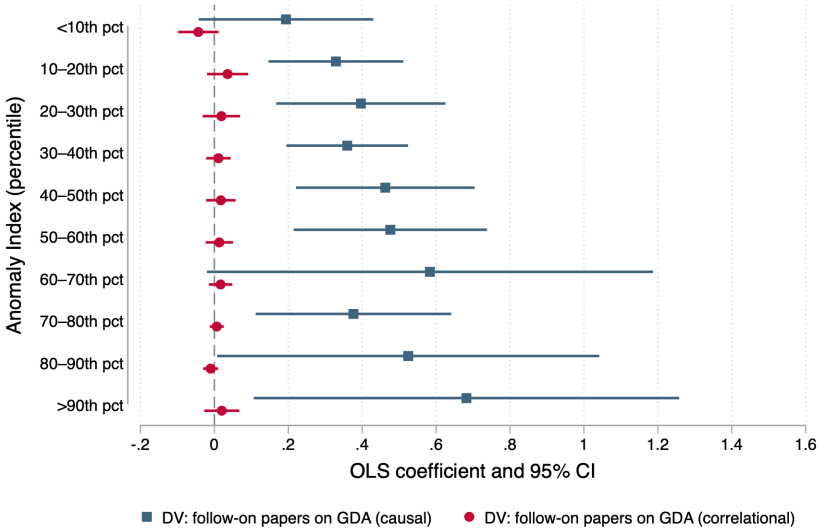


(ii) I(GDA in top 10% DisGeNET Score > 0*)*



Note: The figure examines how the relationship between GWAS and discovery quality varies with the extent of prior genetic knowledge in a disease area. Each bar reports the coefficient on the *GWAS (0/1)* indicator from an OLS regression estimated separately for diseases with limited (below-median) or extensive (above-median) prior genetic knowledge, measured by the number of genes linked to the disease before 2005. The dependent variables indicate whether a newly introduced gene-disease association falls in the bottom half or top decile of the DisGeNET Score distribution. Regressions are estimated at the gene-disease association level and include year and disease fixed effects. Standard errors are two-way clustered by gene and disease.

Figure G.3: More anomalous GWAS discoveries attract proportionally more follow-on research examining causal mechanisms.



Note: The figure examines how the relationship between GWAS discovery and follow-on research varies with the anomalousness of a new gene-disease association. Associations are divided into deciles of the Anomaly Index. Within each decile, I estimate separate cross-sectional OLS regressions of follow-on publications on an indicator for whether the association was introduced by GWAS rather than a candidate gene study. The dependent variables are the number of subsequent articles investigating causal and correlational relationships, respectively. Standard errors are clustered at the gene and disease levels.

Table G.1: Robustness check: GWAS introduce associations involving genes that received fewer publications or were discovered later, relative to candidate gene studies.

	Publications on the gene (pre-2005)		Years since the discovery of the gene (as of 2005)	
	(1)	(2)	(3)	(4)
GWAS (0/1)	-50.84***	-29.44***	-2.978***	-2.126***
	(5.789)	(6.121)	(0.206)	(0.241)
Disease FE	YES	YES	YES	YES
Year of discovery FE	YES	YES	YES	YES
Principal investigator FE	NO	YES	NO	YES
Observations	369,302	348,921	364,661	344,416
Number of diseases	9,863	9,504	9,786	9,422
Mean of the DV	100.57	100.57	12.81	12.81

Note: *, **,*** denote significance at the 5%, 1%, and 0.1% level, respectively. Cross-sectional regressions at the gene-disease association (GDA) level. Std. err. clustered at the gene and disease level. *Publications on the gene (pre-2005)*: count of articles published before 2005 that study the gene in the newly reported gene–disease association; *Years since the discovery of the gene (as of 2005)*: years between the gene's discovery year and 2005 for the gene in the newly reported association; *GWAS*: 0/1 = 1 for new gene-disease associations introduced by a GWAS.

Table G.2: Robustness check: GWAS introduce more gene-disease associations in the lower tail of scientific quality relative to candidate gene studies.

GDA in...	Bottom 45% DisGeNET Score		Bottom 55% DisGeNET Score		Bottom 60% DisGeNET Score	
	(1)	(2)	(3)	(4)	(5)	(6)
GWAS (0/1)	0.0565***	0.0321*	0.104***	0.0705***	0.0727***	0.0434*
	(0.0136)	(0.0163)	(0.0160)	(0.0179)	(0.0153)	(0.0175)
Disease FE	YES	YES	YES	YES	YES	YES
Year of discovery FE	YES	YES	YES	YES	YES	YES
Principal investigator FE	NO	YES	NO	YES	NO	YES
Observations	369,302	348,921	369,302	348,921	369,302	348,921
Number of diseases	9,863	9,504	9,863	9,504	9,863	9,504
Mean of the DV	0.450	0.450	0.550	0.550	0.600	0.600

Note: †, *, **,*** denote significance at the 10%, 5%, 1%, and 0.1% level, respectively. Cross-sectional regressions at the gene-disease association (GDA) level. Std. err. clustered at the gene and disease level. *GDA in bottom 45% DisGeNET Score*: 0/1 = 1 if the gene-disease association has a DisGeNET Score below the 45th percentile of the sample; *GDA in bottom 55% DisGeNET Score*: 0/1 = 1 if the gene-disease association has a DisGeNET Score below the 55th percentile of the sample; *GDA in bottom 60% DisGeNET Score*: 0/1 = 1 if the gene-disease association has a DisGeNET Score below the 60th percentile of the sample; *GWAS*: 0/1 = 1 for new gene-disease associations introduced by a GWAS.

Table G.3: Robustness check: GWAS introduce more gene-disease associations in the top tail of scientific quality relative to candidate gene studies.

GDA in...	Top 5% DisGeNET Score		Top 10% DisGeNET Score		Top 20% DisGeNET Score	
	(1)	(2)	(3)	(4)	(5)	(6)
GWAS (0/1)	0.0398**	0.0292*	0.0411**	0.0355*	0.00371*	-0.000461
	(0.0121)	(0.0126)	(0.0140)	(0.0149)	(0.00171)	(0.00260)
Disease FE	YES	YES	YES	YES	YES	YES
Year of discovery FE	YES	YES	YES	YES	YES	YES
Principal investigator FE	NO	YES	NO	YES	NO	YES
Observations	369,302	348,921	369,302	348,921	369,302	348,921
Number of diseases	9,863	9,504	9,863	9,504	9,863	9,504
Mean of the DV	0.050	0.050	0.100	0.100	0.200	0.200

Note: †, *, **,*** denote significance at the 10%, 5%, 1%, and 0.1% level, respectively. Cross-sectional regressions at the gene-disease association (GDA) level. Std. err. clustered at the gene and disease level. *GDA in top 5% DisGeNET Score*: 0/1 = 1 if the gene-disease association has a DisGeNET Score above the 95th percentile of the sample; *GDA in top 10% DisGeNET Score*: 0/1 = 1 if the gene-disease association has a DisGeNET Score above the 90th percentile of the sample; *GDA in top 20% DisGeNET Score*: 0/1 = 1 if the gene-disease association has a DisGeNET Score above the 80th percentile of the sample; *GWAS*: 0/1 = 1 for new gene-disease associations introduced by a GWAS.

Table G.4: Robustness check: The main results are robust to the inclusion of more stringent fixed effect structures.

Panel A: Including Disease × Year Fixed Effects

	DisGeNET Score of the GDA (1)	I(GDA in bottom 50% DisGeNET Score>0) (2)	I(GDA in top 10% DisGeNET Score>0) (3)	Follow-on articles on the GDA (all) (4)	Follow-on articles on the GDA (causal) (5)	Follow-on articles on the GDA (correlational) (6)
GWAS (0/1)	0.0174***	0.0306†	0.0477**	0.743***	0.407**	0.0290
	(0.00465)	(0.0175)	(0.0145)	(0.219)	(0.127)	(0.0177)
Disease × year FE	YES	YES	YES	YES	YES	YES
Principal investigator FE	YES	YES	YES	YES	YES	YES
Observations	330,378	330,378	330,378	330,378	330,378	330,378
Number of diseases	7,891	7,891	7,891	7,891	7,891	7,891
Mean of the DV	0.054	0.500	0.094	0.595	0.212	0.136

Panel B: Including Disease × Year and Disease × PI Fixed Effects

	DisGeNET Score of the GDA (1)	I(GDA in bottom 50% DisGeNET Score>0) (2)	I(GDA in top 10% DisGeNET Score>0) (3)	Follow-on articles on the GDA (all) (4)	Follow-on articles on the GDA (causal) (5)	Follow-on articles on the GDA (correlational) (6)
GWAS (0/1)	0.0215*	0.0269	0.0523*	1.247***	0.708**	0.0450
	(0.00837)	(0.0278)	(0.0255)	(0.353)	(0.217)	(0.0221)
Disease × year FE	YES	YES	YES	YES	YES	YES
Disease × PI FE	YES	YES	YES	YES	YES	YES
Observations	206,256	206,256	206,256	206,256	206,256	206,256
Number of diseases	7,321	7,321	7,321	7,321	7,321	7,321
Mean of the DV	0.057	0.513	0.104	0.557	0.201	0.130

Note: †, *, **,*** denote significance at the 10%, 5%, 1%, and 0.1% level, respectively. Cross-sectional regressions at the gene-disease association (GDA) level. Std. err. clustered at the gene and disease level. *DisGeNET Score of the GDA*: synthetic measure of scientific value of the discovery provided by DisGeNET; *I(GDA in bottom 50% DisGeNET Score>0)*: 0/1 = 1 if the gene-disease association has a DisGeNET Score below the sample median; *I(GDA in top 10% DisGeNET Score>0)*: 0/1 = 1 if the gene-disease association has a DisGeNET Score in the top sample decile; *Follow-on articles on the GDA (all)*: count of all subsequent studies investigating the gene-disease association; *Follow-on articles on the GDA (causal)*: count of subsequent studies investigating causal relationships between the gene and the disease; *Follow-on articles on the GDA (correlational)*: count of subsequent studies investigating correlational relationships between the gene and the disease; *GWAS*: 0/1 = 1 for new gene-disease associations introduced by a GWAS. Follow-on articles are papers directly working on the gene-disease combination, regardless of whether they cite the study that first introduced it.

Table G.5: Robustness check: Results are robust to the inclusion of gene fixed effects.

	DisGeNET Score of the GDA (1)	I(GDA in bottom 50% DisGeNET Score>0) (2)	I(GDA in top 10% DisGeNET Score>0) (3)	Follow-on articles on the GDA (all) (4)	Follow-on articles on the GDA (causal) (5)	Follow-on articles on the GDA (correlational) (6)
GWAS (0/1)	0.0206***	-0.0255	0.0560***	0.978***	0.497***	0.108***
	(0.00415)	(0.0148)	(0.0126)	(0.240)	(0.133)	(0.0208)
Disease FE	YES	YES	YES	YES	YES	YES
Year of discovery FE	YES	YES	YES	YES	YES	YES
Principal investigator FE	YES	YES	YES	YES	YES	YES
Gene FE	YES	YES	YES	YES	YES	YES
Observations	347,366	347,366	347,366	347,366	347,366	347,366
Number of diseases	9,491	9,491	9,491	9,491	9,491	9,491
Mean of the DV	0.056	0.501	0.097	0.591	0.209	0.130

Note: *, **,*** denote significance at the 5%, 1%, and 0.1% level, respectively. Cross-sectional regressions at the gene-disease association (GDA) level. Std. err. clustered at the gene and disease level. *DisGeNET Score of the GDA*: synthetic measure of scientific value of the discovery provided by DisGeNET; *I(GDA in bottom 50% DisGeNET Score>0)*: 0/1 = 1 if the gene-disease association has a DisGeNET Score below the sample median; *I(GDA in top 10% DisGeNET Score>0)*: 0/1 = 1 if the gene-disease association has a DisGeNET Score in the top sample decile; *Follow-on articles on the GDA (all)*: count of all subsequent studies investigating the gene-disease association; *Follow-on articles on the GDA (causal)*: count of subsequent studies investigating causal relationships between the gene and the disease; *Follow-on articles on the GDA (correlational)*: count of subsequent studies investigating correlational relationships between the gene and the disease; *GWAS*: 0/1 = 1 for new gene-disease associations introduced by a GWAS. Follow-on articles are papers directly working on the gene-disease combination, regardless of whether they cite the study that first introduced it.

Table G.6: Robustness check: The increase in follow-on research on GWAS discoveries is robust to controlling for the prestige of the journal where the article appeared.

	Follow-on articles on the GDA (all)		Follow-on articles on the GDA (causal)		Follow-on articles on the GDA (correlational)	
	(1)	(2)	(3)	(4)	(5)	(6)
GWAS (0/1)	0.525**	0.498**	0.282**	0.271*	0.0550***	0.0505**
	(0.187)	(0.186)	(0.109)	(0.109)	(0.0160)	(0.0157)
SJR journal quartile: Q2	-0.132***		-0.0422***		-0.0462***	
	(0.0220)		(0.0104)		(0.00885)	
SJR journal quartile: Q3	-0.126***		-0.0496**		-0.0445**	
	(0.0311)		(0.0174)		(0.0135)	
SJR journal quartile: Q4	-0.0958		-0.0479		-0.0185	
	(0.122)		(0.0523)		(0.0113)	
Inverse journal rank		0.0000326***		0.0000125***		0.00000754*
		(0.00000388)		(0.00000172)		(0.00000306)
Disease FE	YES	YES	YES	YES	YES	YES
Year of discovery FE	YES	YES	YES	YES	YES	YES
Principal investigator FE	YES	YES	YES	YES	YES	YES
Observations	348,534	348,534	348,534	348,534	348,534	348,534
Number of diseases	9,502	9,502	9,502	9,502	9,502	9,502
Mean of the DV	0.583	0.583	0.204	0.204	0.125	0.125

Note: *, **, *** denote significance at the 5%, 1%, and 0.1% level, respectively. Cross-sectional regressions at the gene-disease association (GDA) level. Std. err. clustered at the gene and disease level. *Follow-on articles on the GDA (all)*: count of all subsequent studies investigating the gene-disease association; *Follow-on articles on the GDA (causal)*: count of subsequent studies investigating causal relationships between the gene and the disease; *Follow-on articles on the GDA (correlational)*: count of subsequent studies investigating correlational relationships between the gene and the disease; *GWAS*: 0/1 = 1 for new gene-disease associations introduced by a GWAS. *SJR journal quartile*: quartile of the journal in which the GDA was first reported, based on SCImago Journal Rankings (Q1 is the omitted category); *Inverse journal rank*: continuous measure of the prestige of the journal in which the GDA was first reported. Follow-on articles are papers directly working on the gene-disease combination, regardless of whether they cite the study that first introduced it.

Table G.7: Robustness check: OLS estimates on the IV subsample replicate the main results.

	DisGeNET Score of the GDA (1)	I(GDA in bottom 50% DisGeNET Score>0) (2)	I(GDA in top 10% DisGeNET Score>0) (3)	Follow-on articles on the GDA (all) (4)	Follow-on articles on the GDA (causal) (5)	Follow-on articles on the GDA (correlational) (6)
GWAS (0/1)	0.027***	0.000	0.075**	1.692*	0.800*	0.075
	(0.008)	(0.025)	(0.023)	(0.718)	(0.356)	(0.047)
Disease FE	YES	YES	YES	YES	YES	YES
Year of discovery FE	YES	YES	YES	YES	YES	YES
Principal investigator FE	YES	YES	YES	YES	YES	YES
Observations	69,157	69,157	69,157	69,157	69,157	69,157
Number of diseases	1,565	1,565	1,565	1,565	1,565	1,565
Mean of the DV	0.058	0.499	0.101	0.689	0.246	0.108

Note: *, **, *** denote significance at the 5%, 1%, and 0.1% level, respectively. Cross-sectional regressions at the gene-disease association (GDA) level, restricted to the subsample with available data for the instrumental variable analysis. Std. err. clustered at the gene and disease level. The smaller number of observations relative to Table 4 reflects the more restrictive fixed-effects specification. *DisGeNET Score of the GDA*: synthetic measure of scientific value of the discovery provided by DisGeNET; *I(GDA in bottom 50% DisGeNET Score>0)*: 0/1 = 1 if the gene-disease association has a DisGeNET Score below the sample median; *I(GDA in top 10% DisGeNET Score>0)*: 0/1 = 1 if the gene-disease association has a DisGeNET Score in the top sample decile; *Follow-on articles on the GDA (all)*: count of all subsequent studies investigating the gene-disease association; *Follow-on articles on the GDA (causal)*: count of subsequent studies investigating causal relationships between the gene and the disease; *Follow-on articles on the GDA (correlational)*: count of subsequent studies investigating correlational relationships between the gene and the disease; *GWAS*: 0/1 = 1 for new gene-disease associations introduced by a GWAS. Follow-on articles are papers directly studying a gene-disease combination, regardless of whether they cite the study that first introduced it.

Table G.8: Instrumental variable estimates: GWAS broaden the breadth and anomalousness of discovery relative to candidate gene studies.

	First Stage	Second Stage (2SLS)		
	GWAS (0/1)	I(GDA with never associated gene>0)	I(GDA with recently discovered gene>0)	Anomaly Index
	(1)	(2)	(3)	(4)
IV: Genotyping cost shift-share	-0.0104**			
	(0.00235)			
GWAS (0/1)		0.472***	0.257***	0.355*
		(0.116)	(0.0625)	(0.157)
Kleibergen-Paap F-statistic	19.5			
Disease FE	YES	YES	YES	YES
Year of discovery FE	YES	YES	YES	YES
Observations	88,581	88,581	88,581	88,032
Number of diseases	1,724	1,724	1,724	1,571

Note: *, **,*** denote significance at the 5%, 1%, and 0.1% level, respectively. Cross-sectional regressions at the gene-disease association (GDA) level. Std. err. clustered at the gene and disease class level. Note that the number of observations is smaller because some diseases lack prevalence data from the pre-GWAS period. The first stage regresses the *GWAS* indicator on the instrument *Genotyping cost shift-share*. The second stage uses the fitted values of *GWAS* as the regressor. *I(GDA with never associated gene>0)*: 0/1 = 1 if the new gene-disease association encompasses a gene never associated with a disease before 2005 (the year of the first GWAS); *I(GDA with recently discovered gene>0)*: 0/1 = 1 if the new gene-disease association encompasses a gene discovered after the year 2000 (the year of the Human Genome Project's first draft completion); *Anomaly Index*: a synthetic measure of how unlikely a given gene was to be associated with a disease given the pattern of genes previously associated with that disease up to the year before; *GWAS*: 0/1 = 1 for new gene-disease associations introduced by a GWAS.

Table G.9: Anomalous gene-disease associations are more likely to fall at both extremes of scientific quality, but not more likely to generate follow-on scientific interest.

Panel A: Empirical anomalies and the quality of discovery

	DisGeNET Score of the GDA		I(GDA in bottom 50% DisGeNET Score>0)		I(GDA in top 10% DisGeNET Score>0)	
	(1)	(2)	(3)	(4)	(5)	(6)
Anomaly Index	0.00994***	-0.000259	0.130***	0.111***	0.0455***	0.00952*
	(0.00296)	(0.00174)	(0.0129)	(0.00918)	(0.00746)	(0.00436)
Disease FE	YES	YES	YES	YES	YES	YES
Year of discovery FE	YES	YES	YES	YES	YES	YES
Principal investigator FE	NO	YES	NO	YES	NO	YES
Observations	365,340	345,040	365,340	345,040	365,340	345,040
Number of diseases	8,746	8,450	8,746	8,450	8,746	8,450
Mean of the DV	0.059	0.059	0.500	0.500	0.100	0.100

Panel B: Empirical anomalies and follow-on science

	Follow-on articles on the GDA (all)		Follow-on articles on the GDA (causal)		Follow-on articles on the GDA (correlational)	
	(1)	(2)	(3)	(4)	(5)	(6)
Anomaly Index	-0.149	-0.148	-0.0995*	-0.114	-0.148***	-0.0977***
	(0.154)	(0.158)	(0.0502)	(0.0675)	(0.0308)	(0.0266)
Disease FE	YES	YES	YES	YES	YES	YES
Year of discovery FE	YES	YES	YES	YES	YES	YES
Principal investigator FE	NO	YES	NO	YES	NO	YES
Observations	365,340	345,040	365,340	345,040	365,340	345,040
Number of diseases	8,746	8,450	8,746	8,450	8,746	8,450
Mean of the DV	0.583	0.583	0.204	0.204	0.125	0.125

Note: *, **,*** denote significance at the 5%, 1%, and 0.1% level, respectively. Cross-sectional regressions at the gene-disease association (GDA) level. Std. err. clustered at the gene and disease level. *DisGeNET Score of the GDA*: synthetic measure of scientific value of the discovery provided by DisGeNET; *I(GDA in bottom 50% DisGeNET Score>0)*: 0/1 = 1 if the gene-disease association has a DisGeNET Score below the sample median; *I(GDA in top 10% DisGeNET Score>0)*: 0/1 = 1 if the gene-disease association has a DisGeNET Score in the top sample decile; *Follow-on articles on the GDA (all)*: count of all subsequent studies investigating the gene-disease association; *Follow-on articles on the GDA (causal)*: count of subsequent studies investigating causal relationships between the gene and the disease; *Follow-on articles on the GDA (correlational)*: count of subsequent studies investigating correlational relationships between the gene and the disease; *Anomaly Index*: a synthetic measure of how unlikely a given gene was to be associated with a disease given the pattern of genes previously associated with that disease up to the year before.

Table G.10: GWAS introduce more anomalous gene-disease associations, even after accounting for prior gene knowledge.

Panel A: Baseline estimates

	Anomaly Index of the GDA			
	(1)	(2)	(3)	(4)
GWAS (0/1)	0.0805***	0.0713***	0.00698***	0.00757***
	(0.00765)	(0.00794)	(0.000927)	(0.00171)
Disease FE	YES	YES	YES	YES
Year of discovery FE	YES	YES	YES	YES
Principal investigator FE	NO	YES	NO	YES
Observations	365,340	345,040	359,303	338,984
Number of diseases	8,746	8,450	8,730	8,429
Mean of the DV	0.027	0.027	0.011	0.011

Panel B: Accounting for prior gene knowledge

	Anomaly Index of the GDA			
	(1)	(2)	(3)	(4)
GWAS (0/1)	0.0713***	0.0116*	0.00747*	0.00894**
	(0.00794)	(0.00506)	(0.00320)	(0.00340)
Disease FE	YES	YES	YES	NO
Year of discovery FE	YES	YES	YES	NO
Principal investigator FE	YES	YES	YES	YES
Gene FE	NO	YES	YES	YES
Prior gene-knowledge controls	NO	NO	YES	YES
Disease $\times$ year FE	NO	NO	NO	YES
Observations	345,040	343,487	343,487	326,507
Number of diseases	8,450	8,442	8,442	7,211
Mean of the DV	0.027	0.023	0.023	0.020

Note: *, **, *** denote significance at the 5%, 1%, and 0.1% level, respectively. Cross-sectional regressions at the gene-disease association (GDA) level. Std. err. clustered at the gene and disease level. *Prior gene-knowledge controls*: separate sets of indicators for bins of the number of unique publications on the gene and the number of distinct diseases previously associated with it. Both counts are constructed from DisGeNET records dated before the focal association's discovery year. *Anomaly Index*: a synthetic measure of how unlikely a given gene was to be associated with a disease given the pattern of genes previously associated with that disease up to the year before; *GWAS*: 0/1 = 1 for new gene-disease associations introduced by a GWAS.

Table G.11: Descriptive statistics of the gene-disease-year panel.

	Candidate gene studies						GWAS					
	mean	median	st d	min	max	N	mean	median	st d	min	max	N
Papers about the gene-disease pair	0.047	0	0.387	0	101	4,673,131	0.088	0	0.980	0	88	108,940
. . . investigating causal relationships	0.024	0	0.219	0	72	4,673,131	0.051	0	0.570	0	53	108,940
. . . investigating correlational relationships	0.016	0	0.172	0	56	4,673,131	0.006	0	0.092	0	7	108,940
Articles weighted by scientific citations	2.211	0	35.641	0	11,338	4,673,131	10.269	0	106.084	0	6,983	108,940
Articles weighted by clinical citations	0.035	0	1.058	0	767	4,673,131	0.199	0	2.627	0	285	108,940
Articles weighted by journal SJR	3.979	0	32.567	0	8,672	4,673,131	7.650	0	84.748	0	7,993	108,940
Post publication (0/1)	0.458	0	0.498	0	1	4,673,131	0.337	0	0.473	0	1	108,940
Year	2010	2010	3.742	2004	2016	4,673,131	2010	2010	3.742	2004	2016	108,940

Note: This table presents descriptive statistics for the gene-disease-year level panel. *Papers about the gene-disease pair*= count of yearly papers about a specific gene-disease combination; *Papers about the gene-disease pair investigating causal relationships*= count of yearly papers about a specific gene-disease combination that investigate causal relationships; *Papers about the gene-disease pair investigating correlational relationships*= count of yearly papers about a specific gene-disease combination that investigate correlational relationships; *Articles weighted by scientific citations*: yearly count of all scientific papers investigating a gene-disease pair weighted by the scientific citations they received; *Articles weighted by clinical citations*: yearly count of all scientific papers investigating a gene-disease pair weighted by the clinical citations they received; *Articles weighted by journal SJR*: yearly count of all scientific papers investigating a gene-disease pair weighted by the SCImago Journal Rank (SJR) of the journal in which they were published; *Post publication (0/1)*= 0/1 = 1 in all years after a gene-disease pair is reported by a paper for the first time; *Year*= average year of the observations in the panel. The table reports only data that are effectively used in the empirical estimates, i.e., excluding observations that are dropped by the inclusion of fixed effects.

Table G.12: Difference-in-differences estimates show that GWAS discoveries attract more follow-on research examining causal mechanisms than candidate gene discoveries.

	All papers	Papers about causal relationships	Papers about correlational relationships
	(1)	(2)	(3)
Post × Publication (0/1)	0.0561***	0.0250***	0.0151***
	(0.00139)	(0.000764)	(0.000700)
... × GWAS (0/1)	0.153***	0.0922***	0.00410
	(0.0222)	(0.0138)	(0.00238)
Gene-disease FE	YES	YES	YES
Disease × year FE	YES	YES	YES
Gene × year FE	YES	YES	YES
Observations	4,782,071	4,782,071	4,782,071
Number of diseases	9,853	9,853	9,853

Note: *, **,*** denote significance at the 5%, 1%, and 0.1% level, respectively. Difference-in-differences panel regressions at the gene-disease-year level. Std. err. clustered at the gene and disease level. *All papers*: yearly count of all scientific papers investigating a gene-disease pair; *Papers about causal relationships*: yearly count of all scientific papers investigating causal relationships about a gene-disease pair according to the AI engine of PubTator3 (Wei et al., 2024); *Papers about correlational relationships*: yearly count of all scientific papers investigating correlational relationships about a gene-disease pair according to the AI engine of PubTator3 (Wei et al., 2024); *Post × Publication*: 0/1 = 1 in all years after a gene-disease association is first reported in a study; *GWAS*: 0/1 = 1 for gene-disease associations introduced by a GWAS.

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